

Generalized ionization energies for full Coulomb atoms

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Abstract

We prove the energy part of the generalized ionization conjecture for full nonrelativistic Coulomb atoms with two electron spin states. The energy needed to remove m electrons is asymptotic to $a_{\text{TF}}m^{7/3}$ whenever $m \rightarrow \infty$ and $Z/m \rightarrow \infty$, with the Thomas–Fermi constant in atomic units. We also obtain both conjectured iterated limits.

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1 Introduction

The ionization energy of an atom measures the cost of removing electrons while leaving its nucleus fixed. The first few electrons removed belong to the outer part of the atom, whereas the leading total energy is largely determined by the much denser inner region. Thomas–Fermi theory, introduced by Thomas and Fermi [15, 3], describes the leading large-charge energy. Lieb and Simon established this connection to the many-body Schrödinger theory [6, 7]. Their energy theorem [7, Theorem III.1] concerns terms of order $Z^{7/3}$. Subtracting two such total-energy asymptotics does not control a much smaller ionization energy: the remainder may exceed the difference being sought.

The generalized ionization conjecture asserts that Thomas–Fermi theory nevertheless determines the leading cost of removing many outer electrons, provided their number remains small compared with the nuclear charge. Solovej formulated this statement through upper and lower large- Z limits, followed by a large removed-electron-number limit [14, Section 3, Equation (1)]. We prove its energy part for the full two-spin Coulomb Hamiltonian, in the stronger joint regime in which both the number removed and the ratio of nuclear charge to that number tend to infinity.

Earlier full-model estimates cover different ionization regimes. Seco, Sigal and Solovej bounded the first ionization energy of a neutral atom by $CZ^{20/21}$, using the improvement recorded in their note added in proof [11, Theorem 1 and p. 315]. Ivrii's estimates imply agreement with the magnitude of the Thomas–Fermi chemical potential for successive one-electron ionization energies of positive atoms when $Z^{5/7} = o(Z - N)$ and $Z - N = o(Z)$, where N is the number of electrons [4, Theorem 6.2.1]. These results leave open the joint neutral-to-positive-ion energy asymptotic addressed here.

Uniform excess-charge bounds were proved in Thomas–Fermi–von Weizsäcker theory by Benguria and Lieb [1]; Solovej proved uniform ionization bounds in reduced Hartree–Fock theory [12] and then in unrestricted Hartree–Fock theory [13]. The Thomas–Fermi–von Weizsäcker and Hartree–Fock results concern different variational problems. The passage to arbitrary correlated fermionic states requires additional screening estimates, obtained here through conditional many-body states.

1.1 The theorem and its normalization

In atomic units, let Z be a positive integer and let

$$H_{Z,N} = \sum_{i=1}^N \left(-\frac{1}{2} \Delta_i - \frac{Z}{|x_i|} \right) + \sum_{1 \leq i < l \leq N} \frac{1}{|x_i - x_l|} \quad (1.1)$$

act on $\bigwedge^N L^2(\mathbb{R}^3; \mathbb{C}^2)$. Write $E_Z(N) = \inf \sigma(H_{Z,N})$ and $E_Z(0) = 0$. For integers $Z > m \geq 1$, define

$$I_m(Z) = E_Z(Z - m) - E_Z(Z).$$

The kinetic coefficient of the corresponding Thomas–Fermi functional is $T = \frac{3}{10}(3\pi^2)^{2/3}$. More explicitly, put

$$\mathcal{T}_Z(\rho) = T \int_{\mathbb{R}^3} \rho^{5/3} - Z \int_{\mathbb{R}^3} \frac{\rho(x)}{|x|} dx + \frac{1}{2} \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{\rho(x)\rho(y)}{|x - y|} dx dy,$$

and let $E_Z^{\text{TF}}(N)$ be its infimum over nonnegative densities of mass N and finite displayed terms. Set $I_m^{\text{TF}}(Z) = E_Z^{\text{TF}}(Z - m) - E_Z^{\text{TF}}(Z)$. The constant in the conjecture is characterized by the classical Thomas–Fermi weak-ionization law [2, Section 6, Corollary 5]:

$$\lim_{Z \rightarrow \infty} I_m^{\text{TF}}(Z) = a_{\text{TF}} m^{7/3}, \quad m > 0. \quad (1.2)$$

Our identification of the coefficient in Section 6 recovers this characterization in the present normalization.

Theorem 1.1 (Generalized ionization energy). *For full nonrelativistic Coulomb atoms with two electron spin states,*

$$\frac{I_m(Z)}{m^{7/3}} \longrightarrow a_{\text{TF}} > 0 \quad \text{whenever} \quad m \longrightarrow \infty, \quad \frac{Z}{m} \longrightarrow \infty, \quad (1.3)$$

where $Z > m \geq 1$ are integers. In particular,

$$\lim_{m \rightarrow \infty} \frac{\limsup_{Z \rightarrow \infty} I_m(Z)}{m^{7/3}} = a_{\text{TF}}, \quad (1.4)$$

$$\lim_{m \rightarrow \infty} \frac{\liminf_{Z \rightarrow \infty} I_m(Z)}{m^{7/3}} = a_{\text{TF}}. \quad (1.5)$$

The inner limits are taken over integer Z with m fixed. No convergence of $I_m(Z)$ at a fixed m is asserted or assumed.

The limits in [Equations \(1.4\) and \(1.5\)](#) are precisely the energy part of the generalized ionization conjecture formulated by Solovej [[14](#), Section 3, Equation (1)]. The joint assertion [Equation \(1.3\)](#) strengthens that order of limits. For historical comparison, Solovej established uniform first-ionization upper bounds and sharp generalized radius estimates in unrestricted Hartree–Fock theory [[13](#), Theorems 3.8 and 1.5]; the generalized conjecture and its Hartree–Fock history are discussed in [[14](#), Section 3]. Here the energy is the exact spectral infimum of [Equation \(1.1\)](#), with no restriction to Slater determinants.

For this same full two-spin Coulomb model, the companion [[9](#), Corollary 1.3] also proves $0 < c \leq E_Z(Z-1) - E_Z(Z) \leq C < \infty$ uniformly for every integer $Z \geq 1$. This first-ionization bound does not assert convergence at fixed m and is distinct from the large- m limits above.

1.2 A price per electron

The proof first allows the particle number to vary. For fixed Z , abbreviate $E_n = E_Z(n)$, and introduce the unconstrained energy and the priced minimum

$$E = \inf_{n \geq 0} E_n, \quad P(\lambda) = \min_{n \geq 0} (E_n + \lambda n), \quad \lambda > 0.$$

The minimum exists, as verified in [Section 2](#). Its advantage is local: replacing the exterior electrons by a trial state need not preserve their number. The price accounts for the change exactly. At the length scale $s = \lambda^{-1/4}$, the removed charge is of order s^{-3} . The main intermediate result identifies its coefficient for every minimizing sector, including at prices where several sectors minimize.

Theorem 1.2 (Deficit at a prescribed price). *There is a universal number $q > 0$ such that, for every sequence satisfying*

$$\lambda = s^{-4}, \quad s \longrightarrow 0, \quad Zs^3 \longrightarrow \infty,$$

and every choice of n minimizing $E_n + \lambda n$, one has

$$s^3(Z - n) \longrightarrow q.$$

The number q is the exterior Coulomb charge of the unique classical solution, tending uniformly to zero at infinity, of

$$\Delta F = 4\pi k((F-1)_+)^{3/2} \quad \text{on } \mathbb{R}^3 \setminus \{0\}, \quad k = (5T/3)^{-3/2},$$

that is bounded above and below by positive multiples of $|x|^{-4}$ near zero. The density $k((F-1)_+)^{3/2}$ has bounded support, and $F(x) = q/|x|$ outside that support. Moreover,

$$a_{\text{TF}} = \frac{3}{7}q^{-4/3}.$$

1.3 Analytic precedents and proof structure

The local kinetic comparison combines two classical constructions. Coherent wave packets provide the upper semiclassical energy [[5](#), Section V.A.1]; see also [[13](#), Lemma 8.2]. Neumann cube eigenvalues and their lattice count supply a lower comparison with the same leading coefficient [[7](#), Theorems III.10–III.13]. Their remainders are controlled by local particle counts. This is different from the coarse Lieb–Thirring kinetic inequality [[8](#)], whose positive universal constant is sufficient for preliminary density bounds; the spectral splitting proof used in the companion follows Rumin [[10](#), Theorem 1.4 and Section 3.1].

Screened exterior potentials and outward Thomas–Fermi comparisons are central to Solovej’s Hartree–Fock argument [13, Sections 6, 10, and 12]. Our use of conditional states requires a separate transfer from local quantum energies to conditional fields, while accounting for changes in particle number. The eventual singular profile belongs to the classical Sommerfeld comparison theory [7, Section IV], including the nonspherical comparisons and chemical-potential estimates of [13, Lemma 4.4, Remark 4.5, Theorem 4.6, and Corollary 4.7]. The issue here is to obtain that profile from the correlated quantum state and preserve its positive singularity through conditioning.

We use the localization, screening, and Thomas–Fermi estimates proved in the companion *Uniform excess charge for Coulomb molecules and the outer radius of neutral atoms* [9]. In particular, its uniform neutral-sector bound $E_Z(Z) - E \leq C$ fixes the zero-price integration constant, and its local-field estimates retain explicit control of an arbitrary energy offset [9, Proposition 12.3 and Proposition 4.5]. The required positive-price conditional comparisons and barrier propagation are proved in Sections 3 and 4. Their offset dependence is essential: a minimizing priced state has energy at most $E + Cs^{-7}$, so the allowed offset grows as $s \rightarrow 0$. A fixed-offset conclusion from the neutral-atom theory cannot by itself yield Theorem 1.1.

Section 2 obtains the rough deficit bound and proves that the rescaled exterior electron mass vanishes beyond a fixed radius. In Section 3, noisy observations of the electrons permit local conditional comparisons while controlling the cost of conditioning. A Thomas–Fermi trial state in a small ball is compared with the priced many-body state. A fresh spatial cut records the local particles and leaves a core outside the patch, whose conditional screened potential is harmonic on the patch. Matching this core comparison to the noisy-observation law yields simultaneous upper and lower implications between local density and local potential. Two posterior laws occur in this argument; keeping them distinct is necessary for the Coulomb comparison.

The remaining difficulty is to retain the singular inner charge through successive averaging. Section 4 constructs a positive subsolution at the innermost scale and propagates it through the observation scales. The construction pays for exceptional conditional data with an error density of vanishing expected mass. The weak differential inequality is proved on open neighborhoods of the retained branches before taking their maximum, so no derivative is restricted to an activity interface. Section 5 then selects conditional data, rescales by s , and obtains a decaying solution of the unit-price Thomas–Fermi equation. The barrier prevents loss of the singularity. A maximum-principle argument identifies its Sommerfeld coefficient and proves uniqueness, which forces the same deficit for every minimizing sector.

Finally, Section 6 integrates the slopes of the priced minimum and brackets the desired particle number between two minimizing sectors. Only monotonicity of $E_Z(N)$ is used in this step; convexity in N is not needed. The analogous Thomas–Fermi calculation identifies the coefficient with Equation (1.2). A sequential argument then gives the specified order of limits in Equations (1.4) and (1.5).

2 Priced sectors and an exterior cutoff

The price per electron permits insertion and deletion comparisons without a constraint on the exterior particle number. We first collect the screening estimates that remain applicable when the energy above the unpriced minimum grows. Throughout, write

$$K(x) = \frac{1}{|x|}, \quad V = ZK, \quad \nu = Z\delta_0, \quad D(f) = \frac{1}{2} \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{f(x)f(y)}{|x-y|} dx dy,$$

and use the constants

$$T = \frac{3}{10}(3\pi^2)^{2/3}, \quad k = \left(\frac{5T}{3}\right)^{-3/2}.$$

These are the Thomas–Fermi constants for two spin states and kinetic energy $-\Delta/2$. For a fixed integer Z , abbreviate $E_n = E_Z(n)$, put $E_0 = 0$, and define

$$E = \inf_{n \geq 0} E_n, \quad P(\lambda) = \min_{n \geq 0} (E_n + \lambda n) \quad (\lambda > 0).$$

The existence of this minimum follows below. The asymptotic regime used until [Section 6](#) is

$$\lambda = s^{-4}, \quad s \longrightarrow 0, \quad Zs^3 \longrightarrow \infty. \quad (2.1)$$

Every assertion concerning a minimizing sector applies to every choice of minimizing index at every system in this regime.

If ψ is a normalized finite-sector form-domain vector, we call it an *unshifted δ -state* when its form energy is at most $E + \delta$, where $\delta \geq 0$. Its position law is $|\psi|^2$, with the spin variables summed unless they are recorded; its one-particle density is denoted by ρ_ψ . Expectations may also include the auxiliary random labels specified below. For a real number v , write $v_+ = \max(v, 0)$ and $v_- = \max(-v, 0)$. Constants $C, c > 0$ can change from line to line. Unless a dependence is displayed, they depend only on this fixed model and on the fixed cutoff and kernel conventions.

2.1 Imported estimates and their hypotheses

Lemma 2.1 (Sector and energy estimates). *The sequence E_n is nonincreasing. If $E_n < E_{n-1}$, there is a normalized ground state in sector n , and $n \leq 2Z + 1 \leq 3Z$. There is a largest such index N_* , with $Z \leq N_* \leq 3Z$, and*

$$E = E_{N_*} = E_n \quad (n \geq N_*).$$

There are universal C, Z_0 such that

$$E_Z(Z) \leq E + C \quad (Z \geq Z_0). \quad (2.2)$$

If Ψ is a normalized sector ground state and F is a bounded real symmetric Lipschitz function of the spatial configuration, then

$$\langle F\Psi, H_{Z,n}F\Psi \rangle - E_n \|F\Psi\|_2^2 = \frac{1}{2} \mathbb{E}_\Psi |\nabla F|^2, \quad (2.3)$$

where the left side is interpreted as a quadratic form and the gradient includes all spatial coordinates. Finally, every unshifted δ -state in a sector $n \leq 3Z$ satisfies

$$\int_{\mathbb{R}^3} \rho_\psi^{5/3} \leq C(Z^{7/3} + \delta). \quad (2.4)$$

Proof. The sector assertions and the multiplier identity are [[9](#), Lemmas 2.1 and 2.4–2.6]; the density estimate is [[9](#), Lemma 3.3]. The stronger estimate $E_Z(Z - 1) \leq E + C$ is [[9](#), Proposition 12.3]. Monotonicity gives [Equation \(2.2\)](#). All these results use exactly the Hamiltonian and the two-spin convention fixed here. $\square$

A one-particle square cut consists of nonnegative bounded-gradient Lipschitz functions o, c with $o^2 + c^2 = 1$. Conditional on the positions, assign the particles independently to the out and core groups with probabilities o^2 and c^2 . The *cut data* record the out labels, positions, and spins. For

a finite sequence of cuts, stage labels are recorded as well. Conditional core states are obtained by the corresponding products of cut factors, so antisymmetry is preserved within each group. This is the localization convention of [9, Lemma 3.4].

The local estimate needed for the later patch comparison has a quantitative condition on these cuts. Fix, once and for all,

$$L = 10^5, \quad A = 40, \quad a_y = |y|/L, \quad B_y = B(y, a_y) \quad (y \neq 0).$$

For an unshifted δ -state put

$$m_y = \max(a_y^{-3}, 1) + \sqrt{\delta a_y}, \quad J_y = V(y) - \sum_{|x_i - y| \geq A a_y} \frac{1}{|y - x_i|},$$

$$P_* = \max \left(1, \sup_{y \neq 0} \frac{\mathbb{E}[\#\{i : x_i \in B_y\}^2]}{m_y^2} \right).$$

The number P_* is finite for each finite system, since the count is at most the sector particle number and $m_y \geq 1$.

Lemma 2.2 (Local fields for arbitrary offsets). *For every normalized unshifted δ -state of finite particle number, $P_* \leq C$. Fix $y \neq 0$ and $C_0 \geq 1$. The data X may be trivial. Otherwise, suppose that the initial cut, or finite sequence of cuts (o_h, c_h) , has total out support covered by at most C_0 cells B_z with*

$$C_0^{-1} \leq a_z/a_y \leq C_0, \quad \sum_h (|\nabla o_h|^2 + |\nabla c_h|^2) \leq \sum_z D_z^2 a_z^{-2} \mathbf{1}_{B_z}, \quad \frac{D_z^2}{a_z m_z} \leq C_0 \sqrt{P_*}.$$

Then, for the complete cut data,

$$\|(\mathbb{E}[J_y | X])_+\|_2 \leq C(C_0) \frac{m_y}{a_y}. \quad (2.5)$$

The constants are independent of Z , the sector, the state, y , and δ , apart from the displayed dependence through m_y .

Proof. This is [9, Proposition 4.5]. Its offset is measured from the minimum E over all sectors, just as here; it has no restriction that the offset stay bounded along a sequence. In all applications below we verify the stronger cut condition $D_z^2/(a_z m_z) \leq C_0$. $\square$

Lemma 2.3 (Annular screening estimates). *Let ψ be a normalized unshifted δ -state of finite particle number. For fixed $0 < \alpha < \beta < \infty$ and every $u > 0$,*

$$\|\#\{i : \alpha u \leq |x_i| \leq \beta u\}\|_2 \leq C_{\alpha, \beta} (\max(u^{-3}, 1) + \sqrt{\delta u}). \quad (2.6)$$

Suppose in addition that $n \leq 3Z$, and set

$$B_{u/8}(x) = V(x) - \sum_{|x_i| < u/8} \frac{1}{|x - x_i|}.$$

Let Y be trivial or be the data of a square cut whose out support lies in $\{u/4 \leq |x| \leq 8u\}$ and whose squared gradient sum is at most C/u^2 . There are conditional versions harmonic on $u/4 < |x| < 6u$ for which

$$\left\| \sup_{u/2 \leq |x| \leq 4u} (\mathbb{E}[B_{u/8}(x) | Y])_+ \right\|_2 \leq C \left(\frac{\max(u^{-3}, 1)}{u} + \sqrt{\frac{\delta}{u}} \right). \quad (2.7)$$

The constant in the second estimate may depend on the fixed gradient bound. Both estimates allow every finite δ .

Proof. The count estimate is [9, Corollary 4.9]. Boundary spheres have zero probability because the one-particle density is absolutely continuous, so open or closed annuli give the same estimate. For the field bound, use [9, Lemma 12.1] with inner cutoff $h = u/8$, broad annulus $u/4 < |x| < 6u$, and smaller annulus $u/2 \leq |x| \leq 4u$. The series in that lemma is

$$\frac{\max(u^{-3}, 1) + \sqrt{\delta u}}{u} + \sum_{j \geq 0} \frac{\max((2^j h)^{-3}, 1) + \sqrt{\delta 2^j h}}{\max(u, 2^j h)}.$$

With $h = u/8$, geometric summation bounds it by the right side of Equation (2.7), also when $u < 1$. The permitted cut support and gradient bound satisfy the fixed-annulus hypotheses of that lemma. The conditional harmonic versions follow by integration against the conditional law: for each fixed finite system the inner sources are uniformly separated from the annulus, so differentiation under the integral is valid. No minimizing property is required in either estimate. $\square$

2.2 The rough charge deficit

Lemma 2.4. *The minimum $P(\lambda)$ is attained. Along Equation (2.1), every minimizing index n is eventually nonzero and strictly bound. For a normalized ground state Ψ in that sector, put $D_0 = E_n - E$. Uniformly over all these choices,*

$$|Z - n| \leq C s^{-3}, \quad 0 \leq D_0 \leq C s^{-7}. \quad (2.8)$$

Proof. By Lemma 2.1, E_n is constant for $n \geq N_*$. Since $\lambda > 0$, the priced objective is strictly increasing thereafter, so its minimum is attained. If a minimizing index is positive, comparison with $n - 1$ gives

$$E_{n-1} - E_n \geq \lambda > 0.$$

It is therefore strictly bound, has a ground state, and satisfies $n \leq 3Z$. A normalized exponential one-electron trial of radius Z^{-1} has energy $-Z^2/2$. Moreover,

$$\frac{\lambda}{Z^2} = \frac{s^2}{(Z s^3)^2} \longrightarrow 0.$$

Thus $P(\lambda) < 0$ eventually, excluding the vacuum.

The neutral sector is an admissible competitor. By Equation (2.2),

$$0 \leq D_0 \leq C + s^{-4}(Z - n). \quad (2.9)$$

For a fixed inner configuration, the spherical average of $B_{s/8}$ on $|x| = s$ equals $(Z - \#\{|x_i| < s/8\})/s$. Its expectation is at least $(Z - n)/s$. Apply Equation (2.7) with trivial data and offset D_0 to obtain, for small s ,

$$Z - n \leq C(s^{-3} + \sqrt{D_0 s}).$$

If $d = Z - n \geq 0$ and $z = s^3 d$, then Equation (2.9) gives $D_0 s^7 \leq C s^7 + z$, whence

$$z \leq C(1 + \sqrt{z + C s^7}).$$

This bounds z . If $d < 0$, Equation (2.9) instead gives $d \geq -C s^4$. These two cases prove the deficit estimate, and substitution in Equation (2.9) proves the bound for D_0 . $\square$

2.3 A cutoff at one fixed exterior scale

Lemma 2.5. *There is a sufficiently large fixed R_0 such that, for every sequence of minimizing sectors and corresponding ground states in Equation (2.1),*

$$s^3 \mathbb{E}_{\Psi} \# \{i : |x_i| > R_0 s\} \longrightarrow 0. \quad (2.10)$$

Proof. Choose a smooth $0 \leq \chi \leq 1$ supported in $[1/2, 4]$ and equal to one on $[1, 2]$. At shell scale $u > 0$, write

$$w_i = \chi(|x_i|/u)^2, \quad S_u = \sum_i w_i, \quad m_u = \mathbb{E} S_u, \quad N_u = \mathbb{E} \# \{i : u/2 \leq |x_i| \leq 4u\}.$$

Test the weak eigen-equation with $S_u \Psi$. In its i th summand the other coordinates have energy at least E_{n-1} . Integration by parts gives for the selected-coordinate kinetic term

$$\frac{1}{2} \operatorname{Re} \int \nabla_i \bar{\Psi} \cdot \nabla_i (w_i \Psi) = \frac{1}{2} \int w_i |\nabla_i \Psi|^2 - \frac{1}{4} \mathbb{E} \Delta_i w_i \geq -\frac{1}{4} \mathbb{E} \Delta_i w_i.$$

Retain only the repulsions from particles in $|x_l| < u/8$; the selected particle lies outside that ball whenever $w_i > 0$. Since $E_{n-1} - E_n \geq s^{-4}$, we obtain

$$s^{-4} m_u \leq C u^{-2} N_u + \mathbb{E} \sum_i w_i B_{u/8}(x_i). \quad (2.11)$$

All these tests are bounded symmetric form-domain multipliers for each fixed system.

Suppose $m_u > 0$. Bias the position law by S_u/m_u , equivalently use the normalized state

$$\Psi_u = \sqrt{S_u/m_u} \Psi.$$

The function $\sqrt{S_u}$ is the Euclidean norm of the vector of cutoffs. It is Lipschitz, including at its zeros, and

$$|\nabla \sqrt{S_u}|^2 \leq C u^{-2} \# \{i : u/2 \leq |x_i| \leq 4u\}.$$

The multiplier identity Equation (2.3) therefore makes Ψ_u an unshifted state of offset at most

$$\delta_u = D_0 + C u^{-2} N_u / m_u. \quad (2.12)$$

In both laws take the same auxiliary square cut, full out on $u/2 \leq |x| \leq 4u$, supported in $u/4 \leq |x| \leq 8u$, with gradient bounded by C/u . Its data Y determine S_u and every position with positive weight. Biasing by the Y -measurable variable S_u/m_u does not change the conditional raw law on $S_u > 0$. Consequently the two conditional harmonic fields there are the same. Let

$$M(Y) = \sup_{u/2 \leq |x| \leq 4u} (\mathbb{E}[B_{u/8}(x) \mid Y])_+$$

be this common conditional supremum. Conditioning first on Y gives

$$\mathbb{E} \sum_i w_i B_{u/8}(x_i) \leq \mathbb{E}[S_u M(Y)] = m_u \mathbb{E}_{\Psi_u} M(Y).$$

The equality of conditional fields is used only on positive-bias data; zero-weight summands are omitted. Their harmonic continuity justifies evaluation at the recorded positions. Applying Equation (2.7) to Ψ_u , with Equation (2.12), now yields

$$\mathbb{E} \sum_i w_i B_{u/8}(x_i) \leq C m_u \left(\frac{\max(u^{-3}, 1)}{u} + \sqrt{D_0/u} \right) + C u^{-3/2} \sqrt{m_u N_u}. \quad (2.13)$$

If $m_u = 0$, the resulting shell estimate is immediate as well.

By Equation (2.8), for $u \geq R_0 s$,

$$s^4 \frac{\max(u^{-3}, 1)}{u} \leq R_0^{-4} + s^4, \quad s^4 \sqrt{D_0/u} \leq C \sqrt{s/u} \leq C R_0^{-1/2}.$$

Choose R_0 large and then s small. The terms multiplying m_u in Equation (2.13) can be absorbed in Equation (2.11). As $m_u \leq N_u$, the mixed term is at most $C u^{-3/2} N_u$, and hence

$$m_u \leq C s^4 (u^{-2} + u^{-3/2}) N_u.$$

The annular count bound and Equation (2.8) also give

$$N_u \leq C (u^{-3} + 1 + s^{-7/2} \sqrt{u}).$$

The shells on which the weights equal one, with $u = 2^j R_0 s$, $j \geq 0$, cover the required exterior. Tonelli's theorem and the two preceding inequalities give

$$\begin{aligned} s^3 \mathbb{E} \# \{ |x_i| > R_0 s \} &\leq C s^7 \sum_{j \geq 0} (u^{-2} + u^{-3/2}) (u^{-3} + 1 + s^{-7/2} u^{1/2}) \Big|_{u=2^j R_0 s} \\ &\leq C R_0 (s^2 + s^{5/2} + s^5 + s^{11/2}) \longrightarrow 0. \end{aligned}$$

Every power of u in this sum is negative. Thus the estimate includes arbitrarily distant shells, with the same fixed R_0 . $\square$

3 Conditional density and field comparison at positive price

We now obtain a local Thomas–Fermi relation for a conditional density. The conditioning retains information about the electron configuration while permitting a controlled change of state on any event of appreciable probability. Two different conditional laws enter the argument: observations define the density to be estimated, whereas a fresh spatial cut defines the core used in the energy comparison. We keep these laws separate and match their potentials only after taking expectation.

Throughout this section, Ψ is a normalized ground state in any sector n minimizing $P(\lambda)$, with $\lambda = s^{-4}$ and Equation (2.1) in force. In particular, $n \leq 3Z$ and $D_0 = E_n - E \leq C s^{-7}$ by Lemma 2.4. All statements about sufficiently small scales are uniform in the choice of this minimizing sector.

3.1 Nested observations and their energy cost

Fix $0 < \varepsilon < 1$, to be chosen in Section 4, and define

$$r_0 = \varepsilon Z^{-1/3}, \quad r_j = 2^j r_0, \quad J_* = \max\{j \geq 0 : r_j \leq s\}, \quad \ell_j = r_j^{1.01} \quad (0 \leq j < J_*). \quad (3.1)$$

Eventually $2r_0 \leq s$. At each scale $j < J_*$, observe the unordered array of points $x_i + \ell_j U_{ji}$. All coordinates of all U_{ji} are independent, also independently of the raw configuration, with density

$$\zeta(u) = c_\zeta \exp\left(-\frac{1}{1-u^2}\right) \mathbf{1}_{\{|u|<1\}}.$$

Labels are forgotten separately in each array. Let $\mathcal{F}_j$ be the information in arrays $j, \dots, J_* - 1$, and let $\mathcal{F}_{J_*}$ be trivial. These sigma-fields decrease with j .

Use the constants $L = 10^5$, $A = 40$ and $a_y = |y|/L$ from [Lemma 2.2](#). Fix a real radial function $g \in C_c^\infty(B(0, 1))$ with $\int g^2 = 1$, set $g_b(x) = b^{-3/2}g(x/b)$, and choose

$$w = 10^{-5}, \quad 0 < c_1 < (10L)^{-1}.$$

Writing $d_x = |x|$, define the master packets and conditional fields by

$$\begin{aligned} D_x^b &= \max(d_x, r_0), & t_x &= c_1 D_x^b \min(D_x^b, s)^w, \\ \mathcal{K}_x(y) &= g_{t_x}(y - x)^2, & \mu_j(y) &= \mathbb{E} \left[\sum_i \mathcal{K}_{x_i}(y) \middle| \mathcal{F}_j \right], \\ H'_j(y) &= V(y) - \lambda - (K * \mu_j)(y). \end{aligned} \tag{3.2}$$

The superscript b indicates the lower clamp, not a power. The width function is globally Lipschitz, with $\text{Lip}(t) \leq c_1(1 + w)s^w$, and $t_x \geq c_1 r_0^{1+w}$. Thus, for each fixed system, μ_j is smooth in its evaluation variable, has mass n , and has deterministic bounds on all spatial derivatives. Its Coulomb potential and first derivatives are also continuous and deterministically bounded. These bounds at a fixed system will justify conditional integration; they are not claimed to be uniform in Z .

We use versions obtained by integrating against the conditional law of the raw configuration. Such versions can, for example, be specified by the finite-array likelihoods in [\[9, Lemma 5.1\]](#). They give, simultaneously in space,

$$\mathbb{E}[\mu_j \mid \mathcal{F}_{j+1}] = \mu_{j+1}, \quad \mathbb{E}[H'_j \mid \mathcal{F}_{j+1}] = H'_{j+1}, \quad \Delta H'_j = -4\pi\nu + 4\pi\mu_j. \tag{3.3}$$

Continuity extends identities first obtained on a countable dense set. In particular, H'_j is subharmonic off the nucleus.

Lemma 3.1 (Cost of an observation event). *Let $j < J_*$ and let $A' \in \mathcal{F}_j$ have probability $p > 0$. There is a normalized antisymmetric state $\Psi_{A'}$ in the same sector whose raw law is the raw marginal conditioned on A' and whose form energy satisfies*

$$\langle \Psi_{A'}, H_{Z,n} \Psi_{A'} \rangle \leq E_n + Cr_j^{-2.02} \log^5(e/p). \tag{3.4}$$

The constant is independent of the particle number, the number of observed arrays, and D_0 .

Proof. If $p_{A'}(x)$ is the probability of the observed event conditional on the raw positions, take $\Psi_{A'} = \sqrt{p_{A'}(x)/p} \Psi$. The multiplier is symmetric and preserves the form domain. The first inequality of [\[9, Lemma 5.2\]](#), with $\ell_j = r_j^{1.01}$, is exactly [Equation \(3.4\)](#). Its proof uses the ground-state multiplier identity in the sector n ; it measures the cost above E_n , independently of the offset of E_n from E . $\square$

Accordingly, the unshifted offset of the tilted state is at most $D_0 + Cr_j^{-2.02} \log^5(e/p)$, while its shifted energy is at most

$$P(\lambda) + Cr_j^{-2.02} \log^5(e/p). \tag{3.5}$$

This distinction is what permits the local comparison below for D_0 up to order s^{-7} .

3.2 Statement and preliminary bounds

Proposition 3.2 (Simultaneous inverse comparison). *There is a universal constant C_{cap} with the following property. Fix*

$$M \geq 2, \quad 0 < h_l < h_h < \infty, \quad 0 < \xi < h_l/16.$$

For all sufficiently far systems along Equation (2.1), at every $j < J_*$ there is an event $\mathcal{G}_j \in \mathcal{F}_j$ such that, with $r = r_j$,

$$\mathbb{P}(\mathcal{G}_j^c) \leq Cr^{25}.$$

On $\mathcal{G}_j$, the following assertions hold simultaneously for $r \leq d_y \leq 4Mr$ and $h_l \leq h \leq h_h$:

$$\begin{aligned} d_y^4 H_j'(y) &\leq C_{\text{cap}}, \\ d_y^6 \mu_j(y) > kh^{3/2} &\implies d_y^4 H_j'(y) \geq h - \xi, \\ d_y^6 \mu_j(y) < kh^{3/2} &\implies d_y^4 H_j'(y) \leq h + \xi. \end{aligned} \tag{3.6}$$

The probability constant and the threshold for sufficiently far systems may depend on the displayed fixed parameters and on the fixed packet conventions; C_{cap} is independent of the band and accuracy parameters.

The two implications compare the field with each positive density height from opposite sides. They will yield an approximate Thomas–Fermi relation in Section 5, while permitting the field itself to be arbitrarily negative where the density is small.

We prove the proposition in the rest of this section. The proof uses the arbitrary-offset screening estimates from Section 2 and the energy cost in Lemma 3.1; it does not apply the fixed-offset inverse comparison of [9, Proposition 10.1] with a growing D_0 .

Fix $j < J_*$ and put $r = r_j$. On the enlarged band $r/2 \leq d_y \leq 8Mr$, write

$$a = a_y, \quad t = t_y, \quad \tau = t/a.$$

Uniformly there, for fixed M ,

$$a \asymp_M r, \quad c_M a^w \leq \tau \leq Cs^w. \tag{3.7}$$

If $\mathcal{K}_x(y) \neq 0$, the Lipschitz bound for the widths gives $|x - y| \leq 2t$ and $t_x/t = 1 + O(s^w)$. Changing variables at scale t therefore gives

$$\int \mathcal{K}_z(y) dz = 1 + O(s^w). \tag{3.8}$$

On the local support, the kernel is bounded by Ct^{-3} and is Lipschitz, in either its center or its evaluation variable, with constant Ct^{-4} .

A count event. Except on an event of probability at most r^{40} , the j th observed array has at most $C_M r^{-3}$ entries in the radial band $[r/4, 12Mr]$. To prove this, condition on the event that a sufficiently large such threshold is exceeded, if its probability is greater than r^{40} . Compact noise support forces every compatible raw configuration to have at least that many points in $[r/8, 13Mr]$. By Lemma 3.1 and Equation (2.6), its conditional expected count is at most $C_M r^{-3}$: indeed,

$$\sqrt{D_0 r} \leq Cr^{-3}(r/s)^{7/2} \leq Cr^{-3}, \quad r^{-0.51} \log^{5/2}(e/r^{40}) = o(r^{-3}).$$

This is a contradiction for a sufficiently large fixed threshold. Denote the retained event by $\mathcal{C}_j$. On it every compatible raw configuration has at most $C_M r^{-3}$ centers relevant to the enlarged band. Integration against the observation posterior consequently gives

$$0 \leq \mu_j(y) \leq C_M r^{-6-3w}, \quad |\nabla \mu_j(y)| \leq C_M r^{-7-4w}. \tag{3.9}$$

A universal field cap. At a fixed point y of the enlarged band, condition on $\{d_y^4 H'_j(y) > b_0\}$ if it has probability $p > 0$. Packets centered outside the exclusion ball in J_y do not contain y , by the width bound and the fixed choice of c_1 . Their potentials at y equal the corresponding point potentials by radial averaging. Dropping the other nonnegative packet potentials and using the observation tower therefore bounds the event-mean of $H'_j(y)$ above by the tilted mean of $J_y - \lambda$. The trivial-data case of [Lemma 2.2](#) gives

$$b_0 \leq C + C(d_y/s)^{7/2} - (d_y/s)^4 + C_M r^{2.49} \log^{5/2}(e/p). \quad (3.10)$$

Here $a < 1$ eventually. The polynomial-power difference on the right has a universal upper bound. Choose a universal C' exceeding the entire deterministic bound by one. Solving [Equation \(3.10\)](#) for p and integrating its stretched-exponential tail yields, for every fixed $P > 0$,

$$\mathbb{E}(d_y^4 H'_j(y) - C')_+ \leq C_{P,M} r^P. \quad (3.11)$$

Subharmonicity turns this into a simultaneous bound. For $3r/4 \leq d_y \leq 6Mr$, take a ball of radius $d_y/10$; it lies in the enlarged band. Its mean bounds $H'_j(y)$ above. The contribution of $C' d_z^{-4}$, after multiplication by d_y^4 , is bounded by a universal constant, and the remaining contribution is at most

$$C r^{-3} \int_{\{r/2 \leq d_z \leq 8Mr\}} (d_z^4 H'_j(z) - C')_+ dz.$$

By [Equation \(3.11\)](#) and Markov's inequality, the latter is at most one outside an event of probability $C_M r^{40}$. We have proved the simultaneous cap

$$H'_j(y) \leq C_{\text{cap}} d_y^{-4} \quad (3r/4 \leq d_y \leq 6Mr) \quad (3.12)$$

with a universal C_{cap} .

3.3 A fine patch comparison with the price included

Fix a point y in the original band and a height $h_l/2 \leq h \leq 2h_h$. Suppose that one of the following events, intersected with $\mathcal{C}_j$, has probability $p \geq r^{42}$:

$$\begin{aligned} d_y^6 \mu_j(y) &> k h^{3/2}, & d_y^4 H'_j(y) &< h - \xi/4, \\ d_y^6 \mu_j(y) &< k h^{3/2}, & d_y^4 H'_j(y) &> h + \xi/4. \end{aligned} \quad (3.13)$$

Write Q for the original joint law conditioned on the event. When spatial cuts are introduced, enlarge this law by fresh labels with their ordinary conditional probabilities given the raw positions. Its raw marginal is the state from [Lemma 3.1](#). Set

$$\delta' = C r^{-2.02} \log^5(e/r^{42}), \quad \delta = D_0 + \delta'.$$

Then the raw state has unshifted offset at most δ , shifted offset at most δ' , and

$$m_y \leq C_M a^{-3}, \quad \delta' = o(a^{-6.98}). \quad (3.14)$$

The constants below may depend on the fixed band parameters.

Put $\beta = a^{0.2}$ and $b = \beta a$, with s small enough that $\beta < 1/100$. For each fixed band parameter M , the scales satisfy

$$b \ll \ell_j \ll t \ll a.$$

Here each ratio of consecutive scales tends to zero uniformly in j , also when $r/s \rightarrow 0$, by Equation (3.7). The fine width b makes the semiclassical errors small; the broader master width t absorbs both this smearing and the observation displacement ℓ_j , while $t/a \rightarrow 0$ permits a local field comparison. The patch construction below has three outputs. Its energy comparison controls the Coulomb discrepancy and a negative-field penalty. The former, together with noisy-array matching, recovers the conditional density; the latter permits the high-density implication to pass to expectation. Finally, the two conditional-expectation identities recover the mean potential. We establish these transfers in that order.

Choose a square cut, using the sine and cosine of a smooth angle, that is full out on $B(y, t_0)$, supported out in $B(y, t_0 + b)$, and has gradients bounded by C/b , where $5a \leq t_0 \leq 6a$ is selected below. Retain the out particles satisfying $|x_i - y| < t_0 - 7b$, and call the other out particles deleted. Let n_o denote the total number of out particles and n_{del} the number of deleted particles; thus the retained count is $n_o - n_{\text{del}}$. Then t_0 may be chosen so that

$$\mathbb{E}_Q n_o^2 \leq C_M a^{-6}, \quad \mathbb{E}_Q n_{\text{del}}^2 \leq C_M \beta a^{-6}. \quad (3.15)$$

For every candidate t_0 , the total out count is bounded by the raw count in $B(y, 7a)$, which lies in a fixed annulus about the nucleus with radius comparable to a . The annular count estimate Equation (2.6), with Equation (3.14), bounds the expected square of this encompassing count by $C_M a^{-6}$, uniformly in t_0 . The deleted particles lie in the raw strip $t_0 - 7b \leq |x_i - y| \leq t_0 + b$. Integrating its squared count over $t_0 \in [5a, 6a]$, each ordered pair contributes for a set of lengths at most $8b$. Averaging selects t_0 for the second bound in Equation (3.15), while preserving the first bound.

The fresh conditional core. In the cut marginal of the tilted state, let X record the out labels, positions, and spins. Slice only on these fresh cut data, not additionally on the noisy arrays. Denote the normalized core slice's density by ρ_X^c and its energy, including the price per core electron, by e_X^λ . Then

$$e_X^\lambda \geq P(\lambda), \quad \Phi = V - \lambda - K * \rho_X^c.$$

The localization products preserve antisymmetry within each group, and the normalized core slices are in their sector form domains for almost every datum [9, Lemma 3.4].

There is a nonnegative random variable $F = F(X)$ such that

$$\Phi \leq F \text{ on } B(y, 7a), \quad \|F\|_{L^2(Q)} \leq C_M a^{-4}. \quad (3.16)$$

For this, condition the unshifted field obtained by subtracting only electrons outside $B(y, 20a)$; all those electrons belong to the core. This field is harmonic on $B(y, 10a)$ and bounds Φ above there. At $z \in B(y, 10a)$, the inclusion $B(y, 20a) \subset B(z, Aa_z)$ bounds it above by $\mathbb{E}_Q[J_z | X]$. The cut satisfies Lemma 2.2 uniformly: a bounded number of comparable cells covers its out support, its gradient coefficients obey $D_z^2 \leq C\beta^{-2}$, and $a_z m_z \geq ca^{-2}$, so that $D_z^2/(a_z m_z) \leq Ca^{1.6} \leq C$. That lemma bounds the positive part in $L^2(Q)$ at each target z . Submeans on balls of radius a , followed by Minkowski's integral inequality, give the positive supremum bound in Equation (3.16).

The patch minimizer. On $\Omega = B(y, t_0 - 4b)$ minimize

$$\mathcal{E}_\Phi(f) = T \int f^{5/3} - \int \Phi f + D(f), \quad f \geq 0, \quad \text{supp } f \subset \bar{\Omega}.$$

The field Φ is bounded and harmonic on a neighborhood of $\bar{\Omega}$: the core is excluded from $B(y, t_0)$, and the nucleus is far outside this patch. Coercivity in $L^{5/3}(\Omega)$, weak lower semicontinuity, and strict convexity give a unique minimizer $\rho = \rho_X$. Its Euler equation is

$$W = \Phi - K * \rho, \quad \rho = kW_+^{3/2} \text{ on } \Omega. \quad (3.17)$$

Nonnegative variations give the one-sided condition where $\rho = 0$; variations of either sign on positive density give the equality. Comparison with zero and $D(\rho) \geq (\int \rho)^2/(24a)$ show that

$$\int \rho \leq CaF. \quad (3.18)$$

These minimizers can be chosen measurably in X . One way to see this is to note their continuous dependence in $L^{5/3}$ on a field in $L^{5/2}(\Omega)$: coercivity bounds a converging sequence of minimizers; weak limits minimize the limiting functional by lower semicontinuity and comparison with its minimizer; uniqueness identifies the limit, and convergence of the minimum values forces convergence of the $L^{5/3}$ norms. Uniform convexity gives strong convergence. The Coulomb form is a continuous positive quadratic form on $L^{5/3}(\Omega)$, as used in this argument.

Let ϑ_b be the radial probability kernel from [9, Lemma 3.6], obtained by averaging a cube kernel over translations and rotations. It is bounded by b^{-3} and supported in $B(0, \sqrt{3}b)$. Define

$$\sigma = \sum_{i \text{ retained}} \vartheta_b(\cdot - x_i).$$

Every such smear lies in Ω by the $7b$ retention margin. We claim

$$\mathbb{E}_Q \left[D(\sigma - \rho) + \int W_- \sigma \right] \leq C_M a^{-7+.02}, \quad \int_{B(y, 4a)} \rho_{\text{tilt}}^{5/3} \leq C_M a^{-7}, \quad (3.19)$$

where ρ_{tilt} is the ordinary one-particle density of the tilted state. In particular, it is not the conditional patch density ρ .

Lower and upper energy comparisons. Write (o, c) for the fresh square cut and, for each out-label set $I \subset \{1, \dots, n\}$, set

$$\Psi_I = \prod_{i \in I} o(x_i) \prod_{i \notin I} c(x_i) \Psi_{A'}, \quad T_o = \frac{1}{2} \sum_{I \subset \{1, \dots, n\}} \sum_{i \in I} \|\nabla_i \Psi_I\|_2^2.$$

Thus T_o is the total out kinetic energy in the localized state. Adding price to the exact localization identity of [9, Lemma 3.4] gives

$$\langle \Psi_{A'}, H_{Z,n} \Psi_{A'} \rangle + \lambda n + \text{err}_{\text{loc}} = \mathbb{E}_Q e_X^\lambda + T_o - \mathbb{E}_Q \sum_{i \text{ out}} \Phi(x_i) + \mathbb{E}_Q \sum_{i < l \text{ out}} K(x_i - x_l), \quad (3.20)$$

with $0 \leq \text{err}_{\text{loc}} \leq C_M b^{-2} a^{-3}$. The price has been included in the core energy and in Φ ; there is no localization error associated with it. [9, Lemma 3.6] allows any measurably selected retained subset and gives

$$T_o \geq \mathbb{E}_Q \left[T \int \sigma^{5/3} - Cb^{-2}(n_o^{4/3} + n_o) \right], \quad (3.21)$$

$$\sum_{i < l \text{ out}} K(x_i - x_l) \geq D(\sigma) - Cn_o/b.$$

The leading kinetic coefficient here is exactly T , for two spin states and kinetic energy $-\Delta/2$.

For almost every fixed slice, the corresponding priced upper comparison is

$$P(\lambda) \leq e_X^\lambda + T \int \rho^{5/3} + Cb^{-2} \int \rho - \int \Phi \rho + D(\rho). \quad (3.22)$$

The coherent-packet construction is standard; see [13, Lemma 8.2]. We spell out why the insertion proof of [9, Lemma 3.5] also proves this priced version. Integrate the rank-one matrices of the orbitals $g_b(x-z)e^{ip \cdot x}$, for both spins, over $|p| \leq (3\pi^2 \rho(z))^{1/3}$ with measure $dz dp/(2\pi)^3$. The resulting one-body matrix γ satisfies

$$0 \leq \gamma \leq 1, \quad \text{Tr } \gamma = \int \rho, \quad \rho_\gamma = g_b^2 * \rho,$$

by Plancherel and momentum-ball integration. Its kinetic trace is

$$\text{Tr}(-\Delta/2)\gamma = T \int \rho^{5/3} + Cb^{-2} \int \rho.$$

First truncate its spectral decomposition to finitely many eigenvectors. Occupy these independently with probabilities given by the eigenvalues. Every realization is a finite Slater determinant, with an integer number of electrons, and its mean repulsion is at most the direct energy of the truncated density by the direct-minus-exchange identity. Its support is in the packet region, which lies strictly inside $B(y, t_0)$ and is separated from the core and nucleus. It may therefore be wedged with the core. Every resulting state has shifted energy at least $P(\lambda)$, and the mean price paid by the inserted electrons is $\lambda \text{Tr } \gamma$.

Positive spectral truncations have increasing densities and convergent traces. The potentials are bounded on the packet region for each fixed slice, so all one-body terms converge; direct energies converge by monotone convergence. Radial smearing does not increase D , and harmonic means replace $\int \Phi(g_b^2 * \rho)$ by $\int \Phi \rho$, including the constant price term. Averaging the integer-sector trials and passing to the limit proves Equation (3.22). This reasoning requires neither an integer value of $\int \rho$ nor a measurable choice of a trial determinant as the slice varies.

For retained particles, harmonic means likewise give $\sum_{i \text{ retained}} \Phi(x_i) = \int \Phi \sigma$. The deleted attractions cost at most $F n_{\text{del}}$. Subtract Equation (3.22) in Equation (3.20), use Equations (3.5) and (3.21), and expand the functional about its minimizer. Convexity of the kinetic term leaves $D(\sigma - \rho)$ and the linear term

$$\int \left(\frac{5T}{3} \rho^{2/3} - W \right) (\sigma - \rho) = \int W_- \sigma.$$

The equality uses Equation (3.17): where $\rho > 0$ the coefficient vanishes, and where $\rho = 0$ it equals W_- . By Equations (3.15), (3.16) and (3.18), the resulting bound is

$$\mathbb{E}_Q \left[D(\sigma - \rho) + \int W_- \sigma \right] \leq \delta' + C_M \left(\sqrt{\beta} a^{-7} + b^{-1} a^{-3} + b^{-2} (a^{-4} + a^{-3}) \right). \quad (3.23)$$

All terms are integrable: for example, $\int \rho \leq CaF$ and $\mathcal{E}_\Phi(\rho) \leq 0$ bound both its kinetic and direct terms by CaF^2 in expectation. The exact cut identity also has integrable absolute Coulomb terms by form-domain admissibility.

Since $b = a^{1.2}$, the errors in Equation (3.23), divided by a^{-7} , are bounded by

$$C_M \left(a^{4.98} \log^5(e/a) + a^{1.1} + a^{2.8} + a^{.6} + a^{1.6} \right).$$

This proves the first part of Equation (3.19). For the second, use $e_X^\lambda \geq P(\lambda)$ directly in Equation (3.20). Dropping the nonnegative out repulsion and using $\mathbb{E}_Q F n_o \leq C_M a^{-7}$ gives $T_o \leq C_M a^{-7}$. The coarse kinetic-density inequality of [9, Lemma 3.4], on the full-out ball containing $B(y, 4a)$, proves the remaining assertion.

3.4 Recovering the conditional density from the patch

By Equation (3.17), the patch field satisfies $\Delta W = 4\pi k W_+^{3/2}$ on $B(y, 4a)$. We will use the bounds

$$W \leq C a^{-4} \text{ on } B(y, 3a), \quad \sup_{B(y, 2t)} |W - W(y)| \leq C\tau(a^{-4} + W(y)_-). \quad (3.24)$$

For the first, compare on the ball of radius $R = 4a$ with

$$U(x) = \frac{C R^4}{(R^2 - |x - y|^2)^4}.$$

For large universal C , $\Delta U \leq 4\pi k U^{3/2}$; it diverges at the boundary. Testing with the positive part of $W - U$ proves the bound. For the second, the Coulomb potential of $\rho \mathbf{1}_{B(y, 2a)}$ is $O(a^{-4})$ with gradient $O(a^{-5})$ on that ball, by the first bound and direct integration of the Coulomb kernel. Adding this potential to W leaves a harmonic function bounded above by $C a^{-4}$. Apply the interior gradient estimate to its nonnegative harmonic distance from this upper bound. Its center value is at most $C a^{-4} + W(y)_-$, which proves the oscillation estimate. These comparisons are classical or weak positive-part tests: compact $L^{5/3}$ densities have continuous potentials, so W is continuous.

Define the local averaging operator

$$\mathcal{R}f(y) = \int \mathcal{K}_z(y) f(z) dz.$$

On every compatible path under Q , including its fresh cut labels,

$$|\mathcal{R}\sigma(y) - \mu_j(y)| \leq C_M r^{-3}(\ell_j + b)t^{-4} = o(a^{-6}). \quad (3.25)$$

Here is the reason this comparison uses the original observation posterior correctly. Match any compatible raw configuration to the j th observed multiset. Each matched displacement is at most $\sqrt{3}\ell_j$, so the raw kernel sum is within $C_M r^{-3}\ell_j t^{-4}$ of the sum evaluated at that multiset. The same is true of its posterior average μ_j at these data. Only $C_M r^{-3}$ entries can contribute, by $\mathcal{C}_j$. All relevant raw centers lie in the fully retained region of the fresh cut, since $t/a \rightarrow 0$, $b/a \rightarrow 0$, and $\ell_j/a \rightarrow 0$. Replacing each such center by its fine smear changes the sum by at most $C_M r^{-3} b t^{-4}$. This proves Equation (3.25) without identifying the two posterior laws. The ratios of these errors to a^{-6} are bounded by

$$C_M(a^{.01-4w} + a^{.2-4w}) = o(1).$$

Let

$$G = \{D(\sigma - \rho) \leq a^{-7+.01}\}.$$

By Equation (3.19), $Q(G^c) \leq C_M a^{.01}$. The function $z \mapsto \mathcal{K}_z(y)$ is a compactly supported Lipschitz test with gradient norm at most $C t^{-5/2}$. Coulomb duality [9, Lemma 3.1] therefore gives, on G ,

$$|\mathcal{R}\rho(y) - \mathcal{R}\sigma(y)| \leq C a^{-7/2+.005} t^{-5/2} = o(a^{-6}). \quad (3.26)$$

Indeed the relative gain is at least $a^{.005-(5/2)w}$. All the small errors here are uniform for fixed parameters along Equation (2.1), including scales with arbitrarily small r/s .

On G , the high-density antecedent in Equation (3.13) forces

$$W(y) \geq (h - \xi/16) d_y^{-4}. \quad (3.27)$$

Otherwise Equation (3.24) would imply $W(z) \leq (h - \xi/32) d_y^{-4}$ on the support of the averaging kernel. This remains true for arbitrarily negative $W(y)$: its upper estimate contains the nonpositive

term $-(1 - C\tau)W(y)_-$. The Euler relation, Equation (3.8), and Equations (3.25) and (3.26) would then contradict the high-density antecedent. In the same way the low-density antecedent forces, on G ,

$$W(y) \leq (h + \xi/16)d_y^{-4}. \quad (3.28)$$

The upper bound in Equation (3.24) lets Equation (3.28) pass to expectation with an $o(a^{-4})$ error.

The high case needs a bound on the negative field on G^c . The high antecedent and Equation (3.25) hold on all paths under Q , and give

$$\int_{B(y, 2t)} \sigma \geq ca^{-6}t^3.$$

Using Equation (3.24) on this ball and absorbing $C\tau W(y)_-$ yields

$$W(y)_- \leq C(a^{-6}t^3)^{-1} \int W_- \sigma + C\tau a^{-4}. \quad (3.29)$$

Consequently Equation (3.19) implies

$$\mathbb{E}_Q[\mathbf{1}_{G^c} W(y)_-] = o(a^{-4}),$$

since the ratio of its bound to a^{-4} is at most $C_M a^{02} \tau^{-3} + C\tau \leq C_M a^{02-3w} + Cs^w = o(1)$. Thus Equation (3.27) also passes to expectation with an $o(a^{-4})$ error. This step explicitly controls the negative part; no lower cap on the conditional field is assumed.

3.5 Matching the potentials of the two conditional laws

We adapt the averaged field transfer of [9, Lemma 5.3] to the present priced fields. The preceding patch estimates supply its required small errors; we give the two-law decomposition and estimates explicitly. We prove

$$|\mathbb{E}_Q(H'_j - W)(y)| = o(a^{-4}). \quad (3.30)$$

Write P^{raw} and P^{ms} for the raw and master-smear electron potentials at y , and $P^{\text{ret}}, P^{\text{del}}$ for the raw potentials of retained and deleted out particles. Because the event defining Q is in $\mathcal{F}_j$, the original observation tower gives $\mathbb{E}_Q(K * \mu_j)(y) = \mathbb{E}_Q P^{\text{ms}}$. The core in W instead uses the fresh-cut tower in the tilted raw law. Combining these two identities, and cancelling both nucleus and price, expresses $\mathbb{E}_Q(H'_j - W)(y)$ as the expectation of

$$(P^{\text{raw}} - P^{\text{ms}}) - P^{\text{del}} - (P^{\text{ret}} - K * \sigma)(y) + K * (\rho - \sigma)(y). \quad (3.31)$$

By Hölder's inequality and the ordinary density bound in Equation (3.19), for $0 < u \leq 4a$ the expected raw potential from $B(y, u)$ is at most

$$\int_{B(y, u)} \frac{\rho_{\text{tilt}}(z)}{|z - y|} dz \leq C_M a^{-4} (u/a)^{1/5}. \quad (3.32)$$

Radial smearing never increases the potential of a point charge and leaves it unchanged outside the smear radius. The master-smearing difference in Equation (3.31) can therefore come only from centers within $2t$ of y ; the fine-smearing difference can come only from centers within $\sqrt{3}b$. Equation (3.32) bounds both by $o(a^{-4})$. Deleted centers have distance comparable to a , so Equation (3.15) gives $\mathbb{E}_Q P^{\text{del}} \leq C_M a^{-4} \sqrt{\beta}$.

For the remaining signed potential, truncate the kernel at height t^{-1} . The test $z \mapsto \min(|z - y|^{-1}, t^{-1})$ has homogeneous Sobolev gradient norm $O(t^{-1/2})$. Coulomb duality, or its compactly supported approximation, and Equation (3.19) give expected absolute truncated error at most

$$C_M t^{-1/2} a^{-7/2+.01} = o(a^{-4});$$

the relative gain is $a^{.01} \tau^{-1/2} \leq C_M a^{.01-w/2}$. The truncation loss for ρ is at most $C a^{-6} t^2$ by Equation (3.24). For σ , radial smearing and Equation (3.32) bound it by

$$C_M a^{-4} ((t + \sqrt{3}b)/a)^{1/5}.$$

These also vanish relative to a^{-4} by Equation (3.7). This proves Equation (3.30).

In the high case, Equations (3.27) and (3.29) give $\mathbb{E}_Q W(y) \geq (h - \xi/16) d_y^{-4} - o(a^{-4})$, whereas the failed comparison in Equation (3.13) gives $\mathbb{E}_Q H'_j(y) \leq (h - \xi/4) d_y^{-4}$. Equation (3.30) contradicts these inequalities eventually. The low case follows in the same way from Equation (3.28) and the upper bound on W . We conclude that each fixed-point failure in Equation (3.13), intersected with $\mathcal{C}_j$, has probability less than r^{42} .

3.6 Spatial and height nets

Take a spatial net of mesh at most r^2 in the original band, with $O_M(r^{-3})$ points, and a fixed finite height net in $[h_l/2, 2h_h]$ of mesh at most $\xi/64$. Exclude the fixed-point failures at all net pairs, the complement of $\mathcal{C}_j$, and the enlarged-cap failure. Their total probability is $O_M(r^{39}) + O_M(r^{40})$, in particular at most $C r^{25}$. All these events are $\mathcal{F}_j$ -measurable.

The normalized density $d_y^6 \mu_j(y)$ changes by at most $C_M r^{1-4w} = o(1)$ between a point and its nearest net point, by Equation (3.9). For the field, work on balls of radius a small fixed multiple of r within the enlarged cap band. Adding the Coulomb potential of the local source μ_j makes H'_j harmonic there and bounds that harmonic function above by $C_M r^{-4-3w}$, using Equations (3.9) and (3.12). The positive-harmonic gradient argument used in Equation (3.24) shows that $v(y) = d_y^4 H'_j(y)$ varies over a displacement $O(r^2)$ by at most

$$C_M r^{1-3w} + C_M r v(y)_-. \quad (3.33)$$

Suppose a high-density failure with tolerance ξ occurs at (y, h) . If $v(y) \geq 0$, Equation (3.33) puts the nearby net value below $h - 3\xi/4$ for small s . If $v(y) < 0$, the same upper estimate is bounded by $(1 - C_M r)v(y) + C_M r^{1-3w}$, and gives the identical conclusion since $h \geq h_l > 16\xi$. Choose a height-net value between $h - \xi/4$ and $h - \xi/8$. The small normalized-density variation preserves the strict high antecedent at this lower height, while the field value violates its comparison with tolerance $\xi/4$. This is an excluded net failure.

A low-density failure has $v(y) > h + \xi > 0$. Its nearest net value is above $h + 3\xi/4$ by Equation (3.33). A height-net value between $h + \xi/8$ and $h + \xi/4$ preserves the low antecedent and again yields an excluded failure with tolerance $\xi/4$. Thus both comparisons hold everywhere on the band for every $h_l \leq h \leq h_h$. Together with Equation (3.12), this proves Proposition 3.2.

4 Continuing a lower barrier

The local density–field comparison in Proposition 3.2 does not by itself exclude a limiting field which is too small near the nucleus. We now carry a positive barrier from the nuclear scale to an arbitrary intermediate scale. The construction keeps the actual conditional density inside

the current radius and uses the Thomas–Fermi reaction outside it. Rare failures of the local comparison are retained as an explicit, small source error. This adapts the intermediate-subsolution construction of [9, Proposition 11.2], using the positive-price comparison proved here to allow the growing offset $D_0 = O(s^{-7})$.

Write $d = |y|$ and put

$$f(t) = 4\pi k(t_+)^{3/2}, \quad b_r(y) = Bd^{-4} \left(1 - \frac{r}{8d}\right) \quad (d \geq r), \quad (4.1)$$

where $B > 0$ will be fixed below. Direct differentiation gives

$$\Delta b_r = Bd^{-6} \left(12 - \frac{20r}{8d}\right) \geq \frac{19}{2}Bd^{-6}.$$

Thus $\Delta b_r \geq f(b_r)$ whenever $4\pi k\sqrt{B} \leq 19/2$. If $R = 2r$, then

$$b_r - b_R \geq \gamma R^{-4} \quad (R \leq d \leq 1.1R), \quad \gamma = \frac{B}{16(1.1)^5} > 0. \quad (4.2)$$

Proposition 4.1 (Conditional barrier invariant). *There are fixed positive constants $\varepsilon, B, C_{\text{inv}}$ and $M > 2$ with the following property. For every sufficiently far system in Equation (2.1), use the observations and kernels of Equations (3.1) and (3.2) with this ε . For every $0 \leq j \leq J_*$ there exist $\mathcal{F}_j$ -measurable functions u_j, p_j such that, with $r = r_j$,*

$$\begin{aligned} \Delta u_j &\geq -4\pi\nu + 4\pi \mathbf{1}_{\{d < r\}} \mu_j - 4\pi p_j + \mathbf{1}_{\{d \geq r\}} f(u_j), \\ b_r &\leq u_j \leq C_{\text{inv}} d^{-4} \quad (d \geq r), \\ u_j &= b_r \quad (d > Mr). \end{aligned} \quad (4.3)$$

The first inequality holds in distributions on $\mathbb{R}^3$. For each fixed system, $u_j - V$ is continuous and locally Lipschitz, with deterministic bounds on compact sets; p_j is a nonnegative bounded density supported in $\{d \leq r_j\}$, also with a deterministic bound. Its expected total mass satisfies

$$\sup_{0 \leq j \leq J_*} \mathbb{E} \int p_j \leq C(r_0^{32} + s^{19/2-3w}) = o(1). \quad (4.4)$$

The deterministic fixed-system regularity bounds need not be uniform in Z ; the constants in Equation (4.4) are uniform.

Proof. We use the weak maximum rule in [9, Lemma 5.4]. In its nonsingular form, if finitely many continuous locally H^1 functions satisfy

$$\Delta v_i \geq g(y) + F(y, v_i),$$

where g is locally bounded and F is measurable in y , continuous in its second variable, and locally bounded on bounded ranges, then their maximum satisfies the same inequality with its own value in the reaction. The lemma also allows a common singular term $-4\pi\nu$ when all branches have continuous locally H^1 offsets from V and the reaction vanishes near the origin. These are the precise forms used below.

Initialization. For sufficiently small fixed ε , the initial conditional density satisfies, outside an event of probability at most Cr_0^{35} ,

$$P_0(y) := K * (\mathbf{1}_{\{d < r_0\}} \mu_0)(y) \leq 0.1 Z/r_0 \quad (r_0 \leq |y| \leq 1.1r_0). \quad (4.5)$$

To prove this, fix y in the indicated annulus and consider the $\mathcal{F}_0$ -event $A_y = \{P_0(y) > 0.05Z/r_0\}$. If its probability is greater than r_0^{40} , [Lemma 3.1](#) realizes its raw conditional marginal as a normalized state in sector n with unshifted offset at most

$$D_0 + Cr_0^{-2.02} \log^5(e/r_0^{40}).$$

By [Equation \(2.8\)](#), $D_0 \leq Cs^{-7}$. Since $r_0/s \rightarrow 0$, the displayed offset is at most $Z^{7/3}$ eventually, with ε fixed. The global density estimate of [\[9, Lemma 3.3\]](#) applies: the state is normalized and antisymmetric and $n \leq 3Z$, so its ordinary density ρ_{A_y} obeys $\int \rho_{A_y}^{5/3} \leq CZ^{7/3}$.

Only packet centers in $|x| < 2r_0$ can contribute to the truncated density defining P_0 . The potential of a truncated packet is at most that of the full packet, and radial smearing bounds the latter by $K(y - x)$. The observation tower and Hölder's inequality give

$$\mathbb{E}[P_0(y) \mid A_y] \leq \int_{|x| < 2r_0} \frac{\rho_{A_y}(x)}{|y - x|} dx \leq CZ^{7/5} r_0^{1/5} = C\varepsilon^{6/5} Z/r_0.$$

Choose ε so small that the last coefficient is less than 0.05. This contradicts the event definition. Hence $\mathbb{P}(A_y) \leq r_0^{40}$.

The minimum packet width is $c_1 r_0^{1+w}$. Integrating $|y - z|^{-2}$ against each bounded packet yields

$$\|\nabla P_0\|_\infty \leq CZr_0^{-2-2w}.$$

A spatial net of spacing at most r_0^2 in the annulus has $O(r_0^{-3})$ points. Its union failure probability is $O(r_0^{37})$, and its interpolation error divided by Z/r_0 is $O(r_0^{1-2w})$. This proves [Equation \(4.5\)](#), with the weaker exponent stated there. The good event is measurable because P_0 is continuous in space.

Let G_0 be this good event and define

$$\mu^{\text{in}} = \mathbf{1}_{G_0} \mathbf{1}_{\{d < r_0\}} \mu_0, \quad p_0 = \mathbf{1}_{G_0^c} \mathbf{1}_{\{d < r_0\}} \mu_0.$$

On $d < 2r_0$ consider the branch

$$v = V - K * \mu^{\text{in}} - 0.7Z/r_0 + C_3(Z/r_0)^{3/2} d^2.$$

Fix C_3 so that $6C_3 \geq 4\pi k 2^{3/2}$ and decrease ε if necessary so that $4C_3 \varepsilon^{3/2} \leq 0.05$. Since $Zr_0^3 = \varepsilon^3$, the quadratic term is at most $0.05Z/r_0$ on this ball. On $r_0 \leq d < 2r_0$ we have $v \leq 2Z/r_0$, and therefore

$$\Delta v = -4\pi\nu + 4\pi\mu^{\text{in}} + 6C_3(Z/r_0)^{3/2}, \quad 6C_3(Z/r_0)^{3/2} \geq f(v).$$

On the lower overlap $r_0 \leq d < 1.06r_0$,

$$v \geq (1/1.06 - 0.1 - 0.7)Z/r_0 > 0.14Z/r_0;$$

on G_0^c the same bound follows with the 0.1 term omitted. On $1.9r_0 < d < 2r_0$,

$$v \leq (1/1.9 - 0.7 + 0.05)Z/r_0 < 0.$$

Fix B satisfying the subsolution condition above and $0 < B \leq 0.1\varepsilon^3$. Define u_0 on an open cover by

$$u_0 = \begin{cases} v, & d < 1.06r_0, \\ \max(v, b_{r_0}), & r_0 < d < 2r_0, \\ b_{r_0}, & d > 1.9r_0. \end{cases}$$

The definitions agree on overlaps. Near $d = r_0$ the branch v alone is active and supplies both required source inequalities. Elsewhere the weak maximum rule proves [Equation \(4.3\)](#). The exterior cap holds for $C_{\text{inv}} \geq \max(B, 32\varepsilon^3)$, and

$$\mathbb{E} \int p_0 \leq n\mathbb{P}(G_0^c) \leq Cr_0^{32}.$$

Choice of the remaining constants. Increase C_{inv} to exceed the universal cap C_{cap} in [Proposition 3.2](#). Choose

$$0 < \lambda_2 < \lambda_1 < \gamma/2, \quad M > 2, \quad C_{\text{inv}}M^{-4} < \lambda_2.$$

The constants λ_1, λ_2 are barrier shifts, distinct from the electron price $\lambda = s^{-4}$. Fix $0 < h_l < B/4$ and $h_h > C_{\text{inv}}$, and then choose $e_0, \xi > 0$ so small that

$$\begin{aligned} 2e_0(2M)^4 &< B/4, & 16\xi + 2e_0 &< \lambda_1 - \lambda_2, \\ \xi &< \min(\lambda_2, h_l/16). \end{aligned} \tag{4.6}$$

All these constants are now fixed. The simultaneous comparison [Proposition 3.2](#) applies with exactly these parameters along the regime under consideration.

One outward step. Suppose the invariant holds at $r = r_j$, where $j < J_*$, and set $R = 2r$ and $u = u_j$. On the good event of [Proposition 3.2](#) for the band $r \leq d \leq 4Mr$, use

$$v_1 = u - \lambda_1 R^{-4}, \quad v_2 = H'_j - \lambda_2 R^{-4}.$$

The two branches have complementary roles. On the gained annulus $r \leq d < R$, the old reaction must supply the actual density, so a retained v_1 must satisfy $f(u) \geq 4\pi\mu_j$. Outside R , the field branch already has Laplacian $4\pi\mu_j$, which must instead dominate $f(v_2)$. The opposite inverse implications in [Equation \(3.6\)](#), together with the unequal shifts, give these two inequalities on the neighborhoods used below. On bad data omit v_2 . Define $\tilde{u}$ by

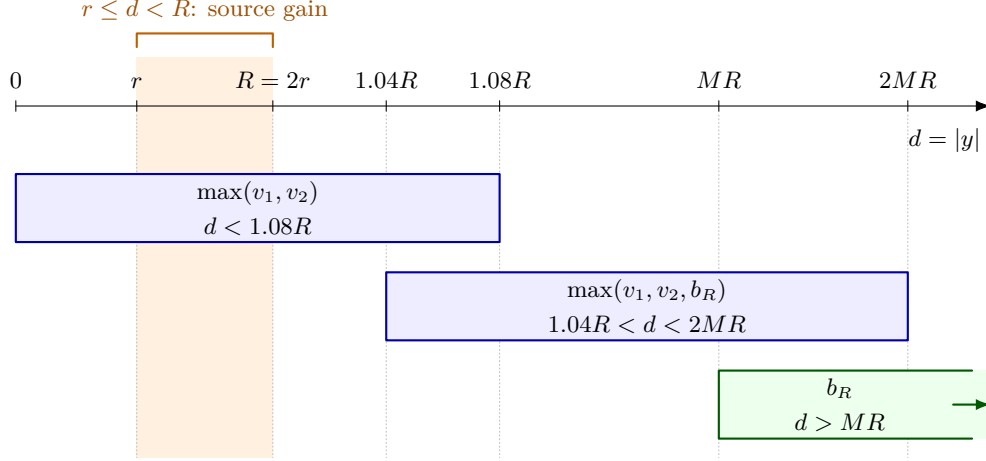
region	$\tilde{u}$
$d < 1.08R$	$\max(v_1, v_2)$
$1.04R < d < 2MR$	$\max(v_1, v_2, b_R)$
$d > MR$	b_R

(4.7)

with the same omission on bad data, and set

$$\tilde{p} = p_j + \mathbf{1}_{\{\text{bad}\}} \mathbf{1}_{\{r \leq d < R\}} \mu_j. \tag{4.8}$$

[Figure 1](#) shows the overlapping regions in [Equation \(4.7\)](#) and the source-gain annulus $r \leq d < R$. On the lower overlap, [Equation \(4.2\)](#) gives $v_1 \geq b_r - \lambda_1 R^{-4} > b_R$. On the upper overlap, the caps and $C_{\text{inv}}M^{-4} < \lambda_2$ put both shifted branches below b_R . Thus the definitions agree. They give $b_R \leq \tilde{u} \leq C_{\text{inv}}d^{-4}$ on $d \geq R$ and $\tilde{u} = b_R$ on $d > MR$; for $R \leq d < 1.08R$ the lower bound already follows from v_1 .



Schematic radial regions, not to scale.
The fields need not be radial.

Figure 1: The overlapping definitions in one continuation step from r to $R = 2r$. The shifted branches are $v_1 = u_j - \lambda_1 R^{-4}$ and $v_2 = H'_j - \lambda_2 R^{-4}$; the second branch is omitted on bad data. The source term changes on the shaded annulus $r \leq d < R$. The barrier gap makes the first two definitions agree on $1.04R < d < 1.08R$, and the caps make the last two agree on $MR < d < 2MR$.

It remains to prove the weak inequality before averaging. We do so on open neighborhoods, to avoid differentiating across the sets where the maximizing branch changes. At a point in one of the open regions in Equation (4.7), retain the candidates whose value is within $e_0 R^{-4}$ of the maximum there. By continuity and finiteness, there is a neighborhood on which every discarded candidate remains strictly below a fixed retained maximizer and every retained candidate v satisfies

$$\tilde{u} - v < 2e_0 R^{-4}. \quad (4.9)$$

The maximum of the retained family is still $\tilde{u}$ throughout this neighborhood. At the origin use the continuous offsets from V to make the same selection.

Every retained branch satisfies the common inequality

$$\Delta v \geq -4\pi\nu + 4\pi\mathbf{1}_{\{d < R\}}\mu_j - 4\pi\tilde{p} + \mathbf{1}_{\{d \geq R\}}f(v) \quad (4.10)$$

on this whole neighborhood. For v_1 , its old inequality suffices on $d < r$, and on $d \geq R$ use $f(u) \geq f(v_1)$. The only new source is on $r \leq d < R$. Bad data there are paid for by Equation (4.8). On good data v_2 is a candidate, so Equation (4.9) gives

$$H'_j \leq u - (\lambda_1 - \lambda_2 - 2e_0)R^{-4}.$$

Moreover $h_l < 7B/8 \leq d^4 u \leq C_{\text{inv}} < h_h$. If $4\pi\mu_j > f(u)$, the high implication of Equation (3.6), at height $h = d^4 u$, would give

$$H'_j \geq u - \xi d^{-4} \geq u - 16\xi R^{-4},$$

contrary to Equation (4.6). Hence $4\pi\mu_j \leq f(u)$, which supplies the required new density.

For v_2 its exact identity $\Delta v_2 = -4\pi\nu + 4\pi\mu_j$ suffices on $d < R$. At a remaining point where it is retained, the lower splice or the included barrier gives $\tilde{u} \geq b_R$ and $d \leq 2MR$. Consequently

$$d^4 v_2 \geq \frac{7B}{8} - 2e_0(2M)^4 > \frac{5B}{8} > h_l, \quad d^4 v_2 \leq C_{\text{cap}} < h_h.$$

If $4\pi\mu_j < f(v_2)$, the low implication of Equation (3.6) would imply

$$H'_j \leq v_2 + \xi d^{-4} \leq v_2 + \xi R^{-4} < v_2 + \lambda_2 R^{-4} = H'_j,$$

a contradiction. Thus $4\pi\mu_j \geq f(v_2)$ there. The branch b_R occurs only outside R and satisfies Equation (4.10) by its subsolution property.

These comparisons concern zero-order densities almost everywhere on the same open neighborhood, including when it crosses $d = R$. The common reaction is $F(y, t) = \mathbf{1}_{\{d \geq R\}} f(t)$. It meets the weak maximum rule's hypotheses; near zero the common V singularity cancels and the reaction vanishes. The rule therefore proves Equation (4.10) for $\tilde{u}$ locally. A partition of unity proves it globally.

Forgetting one observation. Define

$$u_{j+1} = V + \mathbb{E}[\tilde{u} - V \mid \mathcal{F}_{j+1}], \quad p_{j+1} = \mathbb{E}[\tilde{p} \mid \mathcal{F}_{j+1}].$$

We justify the conditional integration of the weak inequality. For a fixed system the positive minimum packet width gives deterministic bounds for μ_j , $K * \mu_j$, and its first derivatives. The initial truncated potential has the same bounds. Thus the initial offsets have deterministic local Lipschitz bounds; subtracting constants, taking finite maxima, and gluing on the fixed open cover preserve such bounds. Where b_R is used, its offset from V is smooth and separated from the origin. The errors remain bounded densities supported in deterministic balls. Conditional integration preserves these bounds. All fields and flags are jointly measurable; the spatial supremum defining the initial flag can be tested on a countable dense set. Fubini's theorem therefore permits conditional integration against each smooth compactly supported test. A countable dense family of nonnegative tests, followed by local approximation, supplies simultaneous almost-sure weak inequalities for all tests.

The conditional tower gives μ_{j+1} as the new inner density, and convexity gives

$$\mathbb{E}[f(\tilde{u}) \mid \mathcal{F}_{j+1}] \geq f(V + \mathbb{E}[\tilde{u} - V \mid \mathcal{F}_{j+1}]).$$

This proves the first line of Equation (4.3) at R . The deterministic lower and upper bounds, the equality beyond MR , and the error support also pass to the conditional average.

Total error. For $r \leq |y| < 2r$, every master packet contributing to $\mu_j(y)$ has center in a fixed annulus comparable to r and width comparable to r^{1+w} . Conditional Jensen, the packet bound, and Equations (2.6) and (2.8) imply

$$\|\mu_j(y)\|_{L^2(\mathbb{P})} \leq Cr^{-3-3w}(r^{-3} + \sqrt{D_0 r}) \leq Cr^{-6-3w},$$

since $r \leq s$ and $D_0 \leq Cs^{-7}$. The bad band event has probability at most Cr^{25} . Cauchy-Schwarz in the data and integration over the gain annulus therefore give

$$\mathbb{E} \int \mathbf{1}_{\{\text{bad}\}} \mathbf{1}_{\{r \leq d < R\}} \mu_j \leq Cr^{25/2+3-6-3w} = Cr^{19/2-3w}.$$

The sum over the geometric sequence of scales is at most $Cs^{19/2-3w}$. Conditional averaging preserves expected total mass. Together with the initial error estimate this proves Equation (4.4) and completes the induction. $\square$

5 The limiting field and its charge

We now prove the charge and profile assertions of [Theorem 1.2](#); [Proposition 6.2](#) will identify the energy coefficient. The barrier is available at every intermediate observation scale, before all observations are forgotten. This lets us select conditional fields which satisfy both the local Thomas–Fermi equation and the small-error barrier.

5.1 Selecting conditional data

Take an arbitrary sequence in [Equation \(2.1\)](#), with an arbitrary minimizing index n in each system. By [Equation \(2.8\)](#), after passing to a subsequence we may suppose

$$q_s := s^3(Z - n) \longrightarrow q_*.$$

Choose integers $l \rightarrow \infty$ sufficiently slowly and a scale $j < J_*$ such that

$$\frac{1}{2l} \leq a_* := \frac{r_j}{s} \leq \frac{1}{l}. \quad (5.1)$$

This is possible because $r_0/s \rightarrow 0$ and the radii are dyadic. In addition to the fixed parameters used in the barrier construction, apply [Proposition 3.2](#) at this scale with $M = l^2$, a fixed upper height greater than $C_{\text{cap}} + 1$, and lower height and tolerance sufficiently small in terms of l . On its good event,

$$\left| \mu_j(y) - k(H'_j(y)_+)^{3/2} \right| \leq l^{-8} |y|^{-6} \quad (r_j \leq |y| \leq ls). \quad (5.2)$$

Indeed, let $u = |y|^4 H'_j(y)$ and $v = (|y|^6 \mu_j(y)/k)^{2/3}$. The cap on u excludes v at or above the upper height: applying the high implication at heights approaching that endpoint would contradict the cap. Above the lower endpoint, the two implications at heights approaching v squeeze u to within the chosen tolerance of v . At or below the lower endpoint they bound u above by that endpoint plus the tolerance. Thus $|u_+ - v| \leq h_l + \xi$, with both u_+ and v in $[0, h_h + 1]$; this uses no lower bound on u . Uniform continuity of $t \mapsto t^{3/2}$ on that interval proves the assertion after choosing h_l and ξ sufficiently small. The spatial range is covered because $4Mr_j \geq 2ls$.

For each fixed l these are fixed parameters of [Proposition 3.2](#); its thresholds hold eventually and its exceptional probability tends to zero. A slow diagonal choice therefore makes [Equation \(5.2\)](#) hold outside a set of probability tending to zero while also ensuring [Equation \(5.1\)](#).

We can select data on which, in addition,

$$\int p_j \leq 1, \quad s^3 \int_{|y| > 2R_0 s} \mu_j(y) dy \leq \eta_s, \quad \eta_s \longrightarrow 0. \quad (5.3)$$

The first excluded probability tends to zero by [Equation \(4.4\)](#). For the second, a master packet centered in $|x| \leq R_0 s$ is contained in $|y| \leq 2R_0 s$ for all sufficiently far systems. The conditional tower and [Equation \(2.10\)](#) show that the expectation of its left side tends to zero, uniformly in j . Markov's inequality supplies a deterministic threshold $\eta_s \rightarrow 0$ with excluded probability tending to zero. These two events and the good comparison event have positive joint probability. Choose a point in their intersection where all the conditional identities and barrier inequalities hold. Only this finite-system selection is needed; every selected density still has total mass exactly n .

5.2 Compactness and the exterior boundary condition

On the selected data, suppress the sequence index and put

$$F_s(x) = s^4(V - K * \mu_j)(sx), \quad \rho_s(x) = s^6 \mu_j(sx).$$

Then

$$\Delta F_s = -4\pi Z s^3 \delta_0 + 4\pi \rho_s, \quad s^4 H'_j(sx) = F_s(x) - 1.$$

The total rescaled electron mass need not be bounded. We instead control the screened field through its spherical mean. Newton's averaging identity and $\int \mu_j = n$ give

$$\overline{F}_s(d) := \frac{1}{4\pi} \int_{\mathbb{S}^2} F_s(d\omega) d\omega = \frac{q_s}{d} + \int_{|x|>d} \left(\frac{1}{d} - \frac{1}{|x|} \right) \rho_s(x) dx. \quad (5.4)$$

In particular $\overline{F}_s(d) \geq q_s/d$.

On every compact annulus, [Proposition 3.2](#) gives the uniform pointwise upper bound $F_s(x) \leq 1 + C_{\text{cap}}|x|^{-4}$ for sufficiently far systems. Together with [Equation \(5.4\)](#) and boundedness of q_s , this bounds F_s in local L^1 away from zero: integrate $|F_s| = 2(F_s)_+ - F_s$ first over spheres and then over radii. [Equation \(5.2\)](#) also bounds ρ_s uniformly on these annuli.

For completeness, choose a ball whose closure avoids the origin, and add to F_s the Coulomb potential of ρ_s restricted to a slightly larger ball. The sum is harmonic on the larger ball. The added potential and its gradient are uniformly bounded by the integrals of $|x - y|^{-1}$ and $|x - y|^{-2}$ against a bounded density. Interior harmonic estimates, using the local L^1 bound, give uniform bounds and equicontinuity on the smaller ball. A diagonal Arzelà–Ascoli argument now gives a subsequence converging locally uniformly on $\mathbb{R}^3 \setminus \{0\}$ to a field F_∞ . The scaled error in [Equation \(5.2\)](#) tends to zero locally uniformly, so the densities converge there to $k((F_\infty - 1)_+)^{3/2}$. Hence

$$\Delta F_\infty = 4\pi k((F_\infty - 1)_+)^{3/2}, \quad F_\infty(x) \leq 1 + C_{\text{cap}}|x|^{-4}. \quad (5.5)$$

The equation holds classically off zero. For example, bounded local sources first give local Hölder regularity of the field; the reaction is then locally Hölder, and interior Poisson regularity gives the asserted conclusion.

The selected tail bound says precisely

$$\int_{|x|>2R_0} \rho_s(x) dx \longrightarrow 0.$$

Local uniform density convergence therefore makes F_∞ harmonic on $|x| > 2R_0$. For $d > 2R_0$, the last term of [Equation \(5.4\)](#) is nonnegative and at most d^{-1} times this tail mass. Consequently

$$\overline{F}_\infty(d) = q_*/d \quad (d > 2R_0). \quad (5.6)$$

These facts imply the uniform boundary condition

$$F_\infty(x) \longrightarrow 0 \quad (|x| \rightarrow \infty). \quad (5.7)$$

Here no positivity has yet been assumed. On large annuli rescaled to a fixed annulus, the upper cap bounds the positive part, and [Equation \(5.6\)](#) bounds the integral of the negative part. Interior harmonic estimates thus give a uniform absolute bound for F_∞ throughout the exterior. Given any sequence $R_i \rightarrow \infty$, its dilates $F_\infty(R_i x)$ have a subsequence converging locally on punctured space to a bounded harmonic function, with the same absolute bound on every punctured compact set. Its singularity at zero is removable, and Liouville's theorem makes it constant. Its spherical mean is zero by [Equation \(5.6\)](#), so the constant is zero. If [Equation \(5.7\)](#) failed, dilating at a sequence of offending radii and taking a convergent subsequence of the corresponding unit directions would contradict this conclusion. This proves uniform decay.

5.3 Transferring the positive barrier

We claim that

$$F_\infty(x) \geq B|x|^{-4} \quad (x \neq 0). \quad (5.8)$$

The field in this assertion is the field without the subtracted price. This distinction permits the reaction comparison below.

Fix an outer radius $S > 0$ and write $U_s(x) = s^4 u_j(sx)$. Inside a_* , the distributional inequality for U_s and the equation for F_s contain the same point-nucleus and density ρ_s terms; the inequality also contains the rescaled error from p_j . Outside a_* , Equation (5.2) gives, once $l > S$,

$$\rho_s(x) \leq k((F_s(x) - 1)_+)^{3/2} + l^{-8}|x|^{-6}.$$

Define the nonnegative correction potential

$$L_s = K * (s^6 p_j(s \cdot) + l^{-8} |\cdot|^{-6} \mathbf{1}_{\{a_* \leq |\cdot| \leq S\}}). \quad (5.9)$$

Scaling Equation (4.3), subtracting the equation for F_s , and using $\Delta K = -4\pi\delta_0$ gives on $B(0, S)$

$$\Delta(U_s - F_s - L_s) \geq \mathbf{1}_{\{|x| \geq a_*\}} [f(U_s) - f(F_s - 1)]. \quad (5.10)$$

In particular, subtracting L_s cancels both the accumulated source error and the approximation error with the required sign.

Choose any

$$c > \left(C_{\text{inv}} S^{-4} - \min_{|x|=S} F_\infty(x) \right)_+.$$

For sufficiently far systems, local uniform convergence on this sphere, the upper bound for U_s , and $L_s \geq 0$ give $U_s - F_s - L_s \leq c$ on the boundary. The same inequality holds throughout the ball. Indeed, test Equation (5.10) with $(U_s - F_s - L_s - c)_+$. On its positive set, $U_s > F_s + L_s + c > F_s - 1$, so the reaction difference is nonnegative. The test yields

$$\int_{B(0, S)} |\nabla(U_s - F_s - L_s - c)_+|^2 \leq 0.$$

It is legitimate on the full ball: the nuclear singularities in $U_s - F_s$ cancel, the remaining offsets are continuous and locally H^1 , and the correction is the potential of a bounded compactly supported density for each fixed system. After cancellation all right-hand sides are locally bounded, including near zero, because the reaction term vanishes on $|x| < a_*$. Approximation therefore allows the compactly supported nonnegative H^1 test.

For each fixed $x \neq 0$, $L_s(x) \rightarrow 0$. The first density in Equation (5.9) has mass at most s^3 by Equation (5.3) and is supported in $|x| \leq a_*$. The second has total mass at most

$$Cl^{-8}a_*^{-3} \leq Cl^{-5},$$

by Equation (5.1); its density near a fixed nonzero evaluation point is $O_x(l^{-8})$. Splitting its potential integral into a small ball around that point and its complement proves the claim. Finally, for fixed $0 < |x| < S$ and sufficiently far systems,

$$U_s(x) \geq B|x|^{-4} \left(1 - \frac{a_*}{8|x|} \right).$$

Pass to the limit in $U_s \leq F_s + L_s + c$, then decrease c to its permitted endpoint. Letting $S \rightarrow \infty$ and using Equation (5.7) proves Equation (5.8).

5.4 The singular profile and uniqueness

The dilation argument below is a direct form of the Sommerfeld comparison principle. The homogeneous zero-price profiles are covered by [13, Lemma 4.4 and Remark 4.5], including nonspherical fields. Here dilation removes the unit price near the singularity; a separate comparison then proves uniqueness for the priced equation.

Lemma 5.1 (Unique decaying singular profile). *There is at most one classical solution F of*

$$\Delta F = 4\pi k((F - 1)_+)^{3/2} \quad \text{on } \mathbb{R}^3 \setminus \{0\}$$

which tends uniformly to zero at infinity and satisfies $c|x|^{-4} \leq F(x) \leq C|x|^{-4}$ near zero for some $c, C > 0$. Every such solution has the uniform singular asymptotic

$$|x|^4 F(x) \longrightarrow A_S := \left(\frac{12}{4\pi k}\right)^2 \quad (|x| \downarrow 0). \quad (5.11)$$

If a solution exists, it is radial.

Proof. Let a and b be the lower and upper limits of $|x|^4 F(x)$ at zero; the hypotheses give $0 < a \leq b < \infty$. Choose points approaching the upper limit and put t equal to their radii. The dilates $u_t(x) = t^4 F(tx)$ satisfy

$$\Delta u_t = 4\pi k((u_t - t^4)_+)^{3/2}.$$

They are bounded above and below by positive multiples of $|x|^{-4}$ on compact annuli. The local potential and harmonic compactness argument used above supplies a subsequential classical limit u with $\Delta u = 4\pi k u^{3/2}$. The definition of b gives $u(x) \leq b|x|^{-4}$ for every $x \neq 0$. After taking a subsequence of the unit directions, equality holds at a unit point. The difference $u - b|x|^{-4}$ has a local maximum there, so

$$4\pi k b^{3/2} \leq 12b.$$

The analogous dilation at points approaching the lower limit touches $a|x|^{-4}$ from above at a unit point and gives $4\pi k a^{3/2} \geq 12a$. Since both constants are positive,

$$b \leq \left(\frac{12}{4\pi k}\right)^2 \leq a.$$

This proves Equation (5.11). The contact argument does not assume radially.

Let F_1, F_2 be two such solutions. For $\alpha > 1$ and $\delta > 0$, the function $G = \alpha F_1 + \delta$ is an upper comparison function:

$$\Delta G = \alpha 4\pi k((F_1 - 1)_+)^{3/2} \leq 4\pi k((G - 1)_+)^{3/2}.$$

If $F_1 \leq 1$ the inequality is immediate; otherwise use $G - 1 \geq \alpha(F_1 - 1)$ and $\alpha^{3/2} \geq \alpha$. The common asymptotic coefficient makes G dominate F_2 on all sufficiently small spheres. Uniform decay and $\delta > 0$ give domination on all sufficiently large spheres. On the intervening annulus, testing with $(F_2 - G)_+$ and using the nondecreasing reaction proves $F_2 \leq G$. Thus this inequality holds everywhere. Let $\delta \downarrow 0$ and $\alpha \downarrow 1$, and interchange the two solutions to obtain equality. Rotations preserve the equation and both boundary conditions, so uniqueness also gives radially. $\square$

The charge assertion of Theorem 1.2. The field F_∞ constructed above satisfies the hypotheses of Lemma 5.1, by Equations (5.5), (5.7) and (5.8). In particular a solution exists, is unique, and is radial. Its exterior mean determines q_* uniquely by Equation (5.6). Moreover, at any fixed $d > 2R_0$,

$$\frac{q_*}{d} = \overline{F}_\infty(d) \geq Bd^{-4} > 0.$$

Denote this uniquely determined positive number by q . The reaction density vanishes beyond $2R_0$, and radially together with Equation (5.6) gives $F_\infty(x) = q/|x|$ there.

Every sequence of systems and every choice of minimizing indices has bounded scaled deficits by Equation (2.8). From any subsequence on which these deficits converge, the preceding construction produces the same unique profile and hence the same limit q . Therefore all scaled deficits converge to q :

$$s^3(Z - n) \longrightarrow q > 0 \quad \text{along Equation (2.1).}$$

The argument allowed an arbitrary minimizing index at every stage, as required. The remaining coefficient identity in Theorem 1.2 is proved in Proposition 6.2. $\square$

6 Fixed particle numbers and the Thomas–Fermi coefficient

We first recover fixed-sector energy differences from the priced deficit proved in Section 5. This step uses monotonicity of the sector energies, without requiring convexity or requiring every sector to minimize at some price. The priced minimum is concave in the price, and its minimizing sectors bound its secant slopes even when the minimizing sector changes. Integrating the deficit law therefore determines the energy at those sectors; monotonicity then brackets the desired particle number. We identify the coefficient by the corresponding Thomas–Fermi calculation.

6.1 Integrating the price and bracketing the sector

Proposition 6.1. *Let q be the deficit coefficient in Theorem 1.2. For every sequence of integers $Z > m \geq 1$ such that $m \rightarrow \infty$ and $Z/m \rightarrow \infty$,*

$$\frac{E_Z(Z - m) - E_Z(Z)}{m^{7/3}} \longrightarrow \frac{3}{7}q^{-4/3}. \quad (6.1)$$

Proof. Fix such a sequence. At each fixed $t > 0$, choose any index n_t minimizing $E_n + tm^{4/3}n$, and put

$$d_{Z,m}(t) = \frac{Z - n_t}{m}, \quad G_{Z,m}(t) = m^{-7/3}[P(tm^{4/3}) - tm^{4/3}Z - E_Z(Z)].$$

The choice $s = t^{-1/4}m^{-1/3}$ satisfies Equation (2.1), since $Zs^3 = t^{-3/4}Z/m \rightarrow \infty$. The deficit assertion of Theorem 1.2 therefore gives

$$d_{Z,m}(t) \longrightarrow qt^{3/4} \quad \text{for every fixed } t > 0, \quad (6.2)$$

for every choice of the minimizing indices.

For $0 < v < t$, the minimizers at the two endpoint prices give

$$-d_{Z,m}(t) \leq \frac{G_{Z,m}(t) - G_{Z,m}(v)}{t - v} \leq -d_{Z,m}(v). \quad (6.3)$$

Indeed, evaluating the objective at n_v gives the upper inequality, and evaluating the objective at n_t at price v gives the lower one. These inequalities do not require differentiability at a price with more than one minimizing sector.

Take a fixed partition $v = t_0 < t_1 < \dots < t_r = t$. Summing Equation (6.3) bounds $G_{Z,m}(t) - G_{Z,m}(v)$ by the two endpoint Riemann sums for the negative deficits. First use Equation (6.2) at the finitely many partition points, and then refine the partition. Since $qt^{3/4}$ is continuous, this proves

$$G_{Z,m}(t) - G_{Z,m}(v) \longrightarrow -\frac{4}{7}q(t^{7/4} - v^{7/4}). \quad (6.4)$$

The behavior at zero price fixes the integration constant. The neutral sector is a competitor, whereas $E_{n_v} \geq E \geq E_Z(Z) - C$ by Equation (2.2); hence

$$-Cm^{-7/3} - v d_{Z,m}(v) \leq G_{Z,m}(v) \leq 0. \quad (6.5)$$

For every fixed $v > 0$ this gives $\limsup |G_{Z,m}(v)| \leq qv^{7/4}$. Combining it with Equation (6.4) and then sending $v \downarrow 0$ yields

$$G_{Z,m}(t) \longrightarrow -\frac{4}{7}qt^{7/4}, \quad \frac{E_{n_t} - E_Z(Z)}{m^{7/3}} = G_{Z,m}(t) + t d_{Z,m}(t) \longrightarrow \frac{3}{7}qt^{7/4}. \quad (6.6)$$

Thus no assertion about the convergence of the priced minimizers as $t \downarrow 0$ was needed.

Choose fixed prices

$$0 < t_- < q^{-4/3} < t_+.$$

By Equation (6.2), their deficits are eventually smaller and larger than m , respectively. Thus $n_{t_-} > Z - m > n_{t_+}$. Monotonicity of the sector energies gives

$$E_{n_{t_-}} - E_Z(Z) \leq E_Z(Z - m) - E_Z(Z) \leq E_{n_{t_+}} - E_Z(Z).$$

Apply Equation (6.6) and let both prices approach $q^{-4/3}$. The two bounds converge to $\frac{3}{7}q(q^{-4/3})^{7/4} = \frac{3}{7}q^{-4/3}$, proving Equation (6.1). $\square$

6.2 The same charge in Thomas–Fermi theory

The classical weak-ionization law was established by B enilan and Brezis [2, Section 6, Corollary 5], following the bounds and chemical-potential integration argument of Lieb and Simon [7, Theorems IV.11–IV.12]. We give the argument in the present normalization to identify its coefficient with the charge obtained from the conditional quantum limit.

We allow real nuclear charges and real electron masses in this subsection. For clarity, the admissible finite-mass class can be written as

$$\mathcal{X} = \{\rho \geq 0 : \rho \in L^1(\mathbb{R}^3) \cap L^{5/3}(\mathbb{R}^3)\}.$$

All terms of $\mathcal{T}_Z$ are finite on this class. For the direct term, interpolation gives $\rho \in L^{6/5}$, and the Sobolev inequality gives the embedding $L^{6/5} \subset (\dot{H}^1)^*$ and the finite Coulomb energy. For the nuclear term, use $|x|^{-1} \in L_{\text{loc}}^{5/2}$ near zero and the finite mass away from zero. Thus this is precisely the finite-mass class in the definition of $E_Z^{\text{TF}}(M)$.

Proposition 6.2. *With the two-spin coefficient T fixed above,*

$$\lim_{Z \rightarrow \infty} [E_Z^{\text{TF}}(Z - 1) - E_Z^{\text{TF}}(Z)] = \frac{3}{7}q^{-4/3}. \quad (6.7)$$

The limit holds through real charges $Z > 1$. Consequently Equation (1.2) holds for every fixed real $m > 0$, with

$$a_{\text{TF}} = \frac{3}{7}q^{-4/3} > 0.$$

Proof. We identify the charge in the unit-price Thomas–Fermi problem, then integrate in the price and use the exact density scaling.

Monotonicity in the electron mass. The energy $E_Z^{\text{TF}}(M)$ is nonincreasing in M . To see this without an attainment assumption, truncate any admissible density of mass M_1 to a large ball. The kinetic, nuclear, and direct terms converge to the original ones. If $M_2 \geq M_1$, place the missing mass in a nonnegative smooth packet of diameter proportional to R , centered at distance R^2 from that ball. For a packet of mass a and this spatial scale, its kinetic and direct energies are $O(a^{5/3}R^{-2})$ and $O(a^2R^{-1})$. Its nuclear term and its interaction with the fixed truncated density tend to zero as $R \rightarrow \infty$. The mass a stays bounded during this construction. Taking first $R \rightarrow \infty$, then the truncation radius to infinity, and then the original energy down to its infimum proves monotonicity. The same argument shows that imposing mass exactly M or at most M gives the same infimum.

The zero-price minimizer is neutral. Apply [9, Lemmas 7.1 and 7.2] to the single positive point source $Z\delta_0$. Their unconstrained problem on $L^{5/3} \cap (\dot{H}^1)^*$ has a unique minimizer ρ_0 , with finite mass at most $2Z$. Its field $\phi_0 = V - K * \rho_0$ satisfies

$$\rho_0 = k(\phi_0)_+^{3/2}, \quad \Delta\phi_0 = -4\pi Z\delta_0 + 4\pi\rho_0, \quad c \min\left(\frac{Z}{|x|}, |x|^{-4}\right) \leq \phi_0(x) \leq C|x|^{-4}. \quad (6.8)$$

The cited lemmas supply the lower bound, finite mass, and decay at infinity. For completeness, the upper bound is independent of Z : on a ball $B(x, R)$ with $R = |x|/2$, compare ϕ_0 with

$$v(y) = \frac{CR^4}{(R^2 - |y - x|^2)^4}.$$

For sufficiently large universal C , direct differentiation gives $\Delta v \leq 4\pi k v^{3/2}$. The function diverges at the boundary, and the positive-part comparison for $\Delta\phi_0 = 4\pi k \phi_0^{3/2}$ in this ball gives $\phi_0(x) \leq CR^{-4}$, as asserted.

Uniqueness makes ρ_0 and ϕ_0 radial. Newton's spherical-mean identity and finite mass give

$$Z - \int \rho_0 = \lim_{d \rightarrow \infty} d\phi_0(d) = 0,$$

where the last equality follows from Equation (6.8). Therefore $\int \rho_0 = Z$ and

$$E_Z^{\text{TF}}(Z) = \min_{\rho \in \mathcal{X}} \mathcal{T}_Z(\rho). \quad (6.9)$$

There is no discrepancy between the variational classes here: every finite-mass admissible density belongs to the cited Coulomb-dual class, and the cited minimizer has finite mass and finite energy terms, hence belongs to $\mathcal{X}$.

The unit-price minimizer. Minimize $\mathcal{T}_Z(\rho) + \int \rho$ over $\mathcal{X}$. Removing density from the region $V \leq 1$ cannot increase this functional: the kinetic term, the contribution $\int (1 - V)\rho$ there, and all removed direct interactions are nonnegative. We may therefore restrict to the ball $B(0, Z)$. Since $V \in L^{5/2}(B(0, Z))$, coercivity and weak lower semicontinuity in $L^{5/3}$ give a minimizer, and strict convexity of the kinetic term gives uniqueness. Denote it by ρ_Z and put $\phi_Z = V - K * \rho_Z$. Nonnegative variations, followed by two-sided multiplicative variations on the positive density, give

$$\rho_Z = k((\phi_Z - 1)_+)^{3/2}, \quad \Delta\phi_Z = -4\pi Z\delta_0 + 4\pi\rho_Z. \quad (6.10)$$

This holds globally: outside $B(0, Z)$ one has $\phi_Z \leq V \leq 1$. The minimizer and its field are radial by uniqueness.

The following comparison is the unit-price case of the classical chemical-potential comparison [13, Lemma 5.1]; we include the short maximum-principle proof. The two fields satisfy

$$\phi_0 \leq \phi_Z \leq \phi_0 + 1. \quad (6.11)$$

For the first inequality, on a positive set of $\phi_0 - \phi_Z - \eta$, with $\eta > 0$, the difference of the two reaction terms in Equations (6.8) and (6.10) is nonnegative. Testing with that positive part gives a nonpositive squared-gradient bound and hence a zero test. The test is compactly supported because both potentials tend to zero at infinity. For the second inequality, $\phi_0 + 1$ solves the same unit-price equation away from the common point source, and the same comparison applies. At the origin the point singularities cancel; the remaining potentials are continuous and locally H^1 , with locally $L^{5/3}$ source densities. Thus the compact tests are justified by H^1 approximation. Let $\eta \downarrow 0$ to obtain Equation (6.11). The continuity and decay used here hold for Coulomb potentials of finite $L^1 \cap L^{5/3}$ densities, as also proved in [9, Lemma 7.1].

Write

$$Q_Z = Z - \int \rho_Z.$$

By Equation (6.11), $\phi_Z \geq 0$ and $\rho_Z \leq k\phi_0^{3/2} \leq C|x|^{-6}$. Newton's identity reads

$$\phi_Z(d) = \frac{Q_Z}{d} + \int_{|y|>d} \left(\frac{1}{d} - \frac{1}{|y|} \right) \rho_Z(y) dy. \quad (6.12)$$

Finite mass gives $Q_Z = \lim_{d \rightarrow \infty} d\phi_Z(d) \geq 0$. At $d = 1$, the nonnegative second term and Equation (6.11) give $Q_Z \leq \phi_Z(1) \leq C + 1$. Using the exterior integral of $C|y|^{-6}$ in Equation (6.12), we conclude

$$0 \leq \phi_Z(d) \leq \frac{C}{d} + Cd^{-4}. \quad (6.13)$$

Choose a fixed sufficiently large R . Then $\phi_Z < 1$ on $d > R$ for every Z , so Equation (6.10) implies $\rho_Z = 0$ there and $\phi_Z(d) = Q_Z/d$.

Identification of the limiting charge. Let $Z \rightarrow \infty$ through real values. On every compact annulus, Equations (6.8) and (6.11) bound ϕ_Z above and below, and Equation (6.10) bounds its source. Local Poisson estimates give a locally uniformly convergent subsequence. Its limit ϕ_∞ satisfies Equation (5.5) and

$$c|x|^{-4} \leq \phi_\infty(x) \leq 1 + C|x|^{-4}.$$

The uniform estimate Equation (6.13) gives decay at infinity. By the uniqueness in Lemma 5.1, this is the same field constructed in Section 5. Its exterior charge is q . At any fixed radius beyond both exterior supports, $Q_Z = d\phi_Z(d) \rightarrow d\phi_\infty(d) = q$. Uniqueness of the subsequential limit therefore proves

$$Q_Z \longrightarrow q \quad (Z \longrightarrow \infty). \quad (6.14)$$

The use of real charges is permitted by the arbitrary-positive-point-source hypotheses of [9, Lemmas 7.1 and 7.2].

Other prices and their energies. For $t > 0$, let $\rho_{Z,t}$ minimize $\mathcal{T}_Z(\rho) + t \int \rho$, and write $Q_Z(t) = Z - \int \rho_{Z,t}$. Existence and uniqueness follow as above, now restricting to $B(0, Z/t)$. With $b = t^{-1/4}$, the transformation

$$\tilde{\rho}(x) = b^6 \rho_{Z,t}(bx), \quad \tilde{\phi}(x) = b^4 (V - K * \rho_{Z,t})(bx)$$

gives the unit-price minimizer and field at nuclear charge $b^3 Z$. Indeed, the transformed functional is exactly

$$\mathcal{T}_{b^3 Z}(\tilde{\rho}) + \int \tilde{\rho} = b^7 \left(\mathcal{T}_Z(\rho_{Z,t}) + t \int \rho_{Z,t} \right),$$

and this transformation is a bijection of admissible densities. Its mass is multiplied by b^3 , so

$$Q_Z(t) = t^{3/4} Q_{t^{-3/4} Z} \longrightarrow q t^{3/4} \quad (t > 0 \text{ fixed}). \quad (6.15)$$

Define

$$\Pi_Z(t) = \min_{\rho \in \mathcal{X}} \left(\mathcal{T}_Z(\rho) + t \int \rho \right), \quad G_Z^{\text{TF}}(t) = \Pi_Z(t) - tZ - E_Z^{\text{TF}}(Z).$$

The endpoint-minimizer comparisons give, for $0 < v < t$,

$$-Q_Z(t) \leq \frac{G_Z^{\text{TF}}(t) - G_Z^{\text{TF}}(v)}{t - v} \leq -Q_Z(v).$$

The exact zero-price identity [Equation \(6.9\)](#) gives

$$-v Q_Z(v) \leq G_Z^{\text{TF}}(v) \leq 0.$$

Consequently the same fixed-partition integration and then $v \downarrow 0$ argument used in [Equations \(6.4\)](#) and [\(6.5\)](#) gives

$$G_Z^{\text{TF}}(t) \longrightarrow -\frac{4}{7} q t^{7/4}, \quad \mathcal{T}_Z(\rho_{Z,t}) - E_Z^{\text{TF}}(Z) \longrightarrow \frac{3}{7} q t^{7/4}. \quad (6.16)$$

Each priced minimizer is also a minimizer at its own mass: an improvement at that mass would improve the priced objective. For fixed $t_- < q^{-4/3} < t_+$, [Equation \(6.15\)](#) places these two masses on opposite sides of $Z - 1$ for all large Z . The mass monotonicity proved above and [Equation \(6.16\)](#), followed by $t_-, t_+ \rightarrow q^{-4/3}$, give [Equation \(6.7\)](#).

Every fixed removed mass. For $m > 0$ the density scaling

$$\rho(x) = m^2 \sigma(m^{1/3} x)$$

multiplies the mass by m and satisfies $\mathcal{T}_Z(\rho) = m^{7/3} \mathcal{T}_{Z/m}(\sigma)$. It is a bijection between the corresponding admissible classes. Hence

$$E_Z^{\text{TF}}(N) = m^{7/3} E_{Z/m}^{\text{TF}}(N/m), \quad I_m^{\text{TF}}(Z) = m^{7/3} I_1^{\text{TF}}(Z/m). \quad (6.17)$$

For fixed $m > 0$, the real charge Z/m tends to infinity with Z . Applying [Equation \(6.7\)](#) proves [Equation \(1.2\)](#) and the claimed value of a_{TF} . This also completes the coefficient assertion of [Theorem 1.2](#). $\square$

6.3 The specified order of limits

Completion of the proof of Theorem 1.1. The joint conclusion [Equation \(1.3\)](#) follows from [Propositions 6.1](#) and [6.2](#). To deduce the iterated limits, suppose first that, along unbounded integers m_j , the normalized inner upper limit exceeds a_{TF} by a fixed $\eta > 0$. By the definition of a limsup, for each j one may choose an arbitrarily large integer $Z_j > m_j$, in particular $Z_j/m_j \geq j$, with

$$\frac{I_{m_j}(Z_j)}{m_j^{7/3}} > a_{\text{TF}} + \eta/2.$$

This contradicts Equation (1.3). The same selection applies if these inner upper limits are infinite. Likewise, a normalized inner lower limit less than $a_{\text{TF}} - \eta$ along unbounded m_j allows choosing $Z_j/m_j \geq j$ with normalized energies below $a_{\text{TF}} - \eta/2$, again a contradiction. Therefore

$$\limsup_{m \rightarrow \infty} \frac{\limsup_{Z \rightarrow \infty} I_m(Z)}{m^{7/3}} \leq a_{\text{TF}}, \quad \liminf_{m \rightarrow \infty} \frac{\liminf_{Z \rightarrow \infty} I_m(Z)}{m^{7/3}} \geq a_{\text{TF}}.$$

Since the inner lower limit never exceeds the inner upper limit, both outer limits exist and equal a_{TF} . The construction always chooses Z only after m is fixed; no convergence in Z at a fixed m has been assumed. $\square$

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